Journal of International Money and Finance 156 (2025) 103342

Contents lists available at ScienceDirect

Journal of International Money and Finance

journal homepage: www.elsevier.com/locate/jimf

The global financial cycle and macroeconomic tail risks ☆

Johannes Beutel [a], Lorenz Emter [b,c], Norbert Metiu [a,*], Esteban Prieto [a,d], Yves Schüler [a]

[a] Deutsche Bundesbank, Frankfurt am Main, Germany
[b] European Central Bank, Frankfurt am Main, Germany
[c] Trinity College Dublin, Dublin, Ireland
[d] Halle Institute for Economic Research (IWH), Halle, Germany

HIGHLIGHTS

- Contractionary shocks to U.S. financial conditions and monetary policy raise down-side risks to growth around the world.
- The left tail of the conditional output growth distribution responds more strongly.
- Effects are particularly strong for countries with less flexible exchange rates, higher foreign currency exposures, and higher levels of private sector leverage.

ARTICLE INFO

JEL classification:
C32
E23
E32
E44
F44

Keywords:
Financial shocks
Monetary policy
Global financial cycle
Growth-at-risk
International spillovers
Quantile VAR

ABSTRACT

We study the link between the global financial cycle and macroeconomic tail risks using quantile vector autoregressions. Contractionary shocks to financial conditions and monetary policy in the United States cause elevated downside risks to growth around the world. By tightening financial conditions globally, these shocks affect the left tail of the conditional output growth distribution more strongly than the center of the distribution. This effect is particularly pronounced for countries with less flexible exchange rate arrangements, higher foreign currency exposures, and higher levels of private sector leverage, suggesting that exchange rate policies and macroprudential policies can mitigate downside risks to growth.

☆ We thank Simone Manganelli, Georgios Georgiadis, Michal Franta, Alan M. Taylor, Felix Thierfelder, and seminar participants at the European Central Bank, the Deutsche Bundesbank, the Central Bank of Ireland, Trinity College Dublin, the EEA 2020, the 4th ESCB Research Cluster 3 Workshop, and the CFS-CMStatistics Conference 2024 for valuable comments and suggestions. The majority of the work on this paper was done while Lorenz Emter was at the Central Bank of Ireland. The views expressed in this paper are those of the authors and do not necessarily represent the views of the Deutsche Bundesbank, the Central Bank of Ireland, or the European Central Bank.

* Corresponding author.
Email addresses: johannes.beutel@bundesbank.de (J. Beutel), lorenz.emter@ecb.europa.eu (L. Emter), norbert.metiu@bundesbank.de (N. Metiu), esteban.prieto@bundesbank.de (E. Prieto), yves.schueler@bundesbank.de (Y. Schüler).

https://doi.org/10.1016/j.jimonfin.2025.103342

Available online 15 April 2025
0261-5606/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

Since the global financial crisis, policymakers have become increasingly concerned with the risk of large negative output growth realizations. As a result, a rapidly growing body of research on downside risks to growth has emerged. Tail risks matter from a theoretical perspective, for instance, in models with agents who are averse to Knightian uncertainty (Ilut and Schneider, 2014), or in a New Keynesian framework with heteroscedastic output gap volatility driven by endogenous financial conditions (Adrian et al., 2020). The empirical literature emphasizes that macroeconomic tail risks increase when domestic financial conditions tighten (e.g., Adrian et al., 2019; Brownlees and Souza, 2021). At the same time, domestic financial conditions are to a substantial degree determined by global forces, as documented in a literature studying global liquidity and the global financial cycle (e.g., Eickmeier et al., 2014; Rey, 2015). Global financial conditions, in turn, are greatly influenced by financial conditions and monetary policy in the United States (e.g., Bruno and Shin, 2015; Miranda-Agrippino and Rey, 2020). Yet, the questions of how unexpected changes in U.S. financial conditions and monetary policy affect macroeconomic tail risks in other countries and which country characteristics increase the vulnerability to such changes have received little attention in the literature.

In this paper, we investigate the effects of shocks to U.S. financial conditions and monetary policy on downside risks to growth around the world. To model the international transmission of U.S. financial and monetary policy shocks to tail risks, we estimate Bayesian quantile vector autoregressions (VARs) on data for 44 advanced and emerging economies, using the excess bond premium (EBP) proposed by Gilchrist and Zakrajsek (2012) as a measure of exogenous changes in U.S. financial conditions and a measure of U.S. monetary policy shocks derived from Miranda-Agrippino and Ricco (2021). This econometric framework allows us to compute *quantile* impulse responses for the median, the upper tail and the lower tail of each country's conditional GDP growth distribution. We then relate the quantile impulse responses to various country characteristics.

Three main results emerge. First, we find that an exogenous tightening in U.S. financial conditions raises macroeconomic tail risks internationally. After four quarters, the effect of the shock on the lower tail is on average four times stronger than on the median of the conditional distribution of GDP growth, while the impact on the upper tail is positive and far less pronounced. Second, we find that an unexpected tightening in U.S. monetary policy also has stronger effects on the lower tail of the conditional GDP growth distribution than on the median and the upper tail. Hence, following U.S. financial and monetary policy shocks, the conditional distribution of GDP growth changes its shape and becomes more negatively skewed as the lower tail widens. Third, we find that certain country characteristics matter significantly for the international transmission of these shocks on the *lower tail* of the conditional GDP growth distribution. Specifically, following a shock to U.S. financial conditions, downside risks to growth increase significantly more for countries with fixed exchange rates, higher foreign currency exposure, and a higher level of household leverage. Following a U.S. monetary policy shock, macroeconomic tail risks increase more in countries with a relatively less flexible exchange rate regime. These characteristics, if at all, only weakly matter for the *median* of the conditional GDP growth distribution.

The results hold up to various robustness checks. Above all, we obtain virtually identical results when using the quantile VAR model of Chavleishvili and Manganelli (2024), in which the contemporaneous relation between the endogenous variables is estimated explicitly. We also find stronger effects on the lower tail than on the median or the upper tail when projecting the quantiles of each country's GDP growth on the EBP in quantile local projections. In addition, the main conclusions continue to hold when using an alternative measure of U.S. financial conditions, controlling for additional U.S. variables, varying the identifying assumptions, and varying the sample period.

Our findings provide a new angle on how shocks that originate in the U.S. affect foreign economies by documenting their impact on macroeconomic *tail* risks. This represents a novel contribution to an extensive literature on the international transmission of U.S. financial shocks and to the literature on international spillovers from U.S. monetary policy, which have focused on the effect of these shocks on the mean.[1] By taking an international perspective, we also bring a new dimension to the literature on downside risks to growth, which has thus far primarily studied the relationship between macro tail risks and *domestic* financial conditions. For instance, Adrian et al. (2019) show that tighter U.S. financial conditions are associated with higher downside risks to U.S. GDP growth.[2] Only few studies have taken an international perspective. Chuliá et al. (2024), for instance, document that deteriorating U.S. financial conditions are linked to a reduction in real credit growth abroad, with particularly pronounced effects at the lower quantiles. In addition, Gaechter et al. (2023) show that structural differences across countries – e.g., in trade openness, financial sector size, public spending, and government effectiveness – give rise to differences in how financial risk indicators affect the projected distribution of future growth outcomes. We contribute to a recent strand of the literature that seeks to move from reduced-form models to modeling the structural shocks that drive downside risks to GDP growth, which has analyzed the domestic relationship between shocks and macroeconomic outcomes (e.g., Loria et al., 2025; Chavleishvili and Manganelli, 2024).

Our paper is related to a long-standing literature on international monetary policy dependence in open economies. The classical "trilemma" (or "impossible trinity") of international macroeconomics postulates that countries face a trade-off among the objectives

[1] On the international transmission of financial shocks, see, e.g., Helbling et al. (2011), Fink and Schüler (2015), Eickmeier and Ng (2015), Abbate et al. (2016), Metiu et al. (2016), Cesa-Bianchi et al. (2018), and Born and Enders (2019). On the international transmission of monetary policy shocks, see, e.g., Kim (2001), Mackowiak (2007), Bruno and Shin (2015), Rey (2015), Fratzscher et al. (2017), Degasperi et al. (2020), and Miranda-Agrippino and Rey (2020). The aforementioned papers study the effect of monetary policy shocks on the mean. In parallel work, Arbatli-Saxegaard et al. (2024) estimate the effect of U.S. monetary policy on the lower tail output growth in cross-country panel quantile regressions. Consistent with our results, they find that downside risks to growth increase after a monetary tightening. Our results provide a detailed assessment of the role of country characteristics in the transmission.

[2] Related papers include: Cecchetti (2008), De Nicoló and Lucchetta (2017), Aikman et al. (2019), Plagborg-Møller et al. (2020), Brownlees and Souza (2021), and Adrian et al. (2022).

of exchange rate stability, free capital mobility, and independent monetary policy (e.g., Mundell, 1960; Fleming, 1962). Considerable evidence suggests that floating exchange rates enable countries with an open capital account to pursue an independent monetary policy (e.g., Obstfeld et al., 2005; Klein and Shambaugh, 2015; Bekaert and Mehl, 2019), and that emerging market economies with flexible exchange rates are better able to use their own monetary policies to dampen the influence of foreign financial and monetary developments on domestic financial variables (see, e.g., Obstfeld et al., 2019; Obstfeld, 2021). By contrast, Rey (2015) argues that the scale of financial integration in recent decades has turned the trilemma into a dilemma in which, regardless of the exchange rate regime, national monetary policies are constrained by global co-movement in gross capital flows, banking sector leverage, credit, and asset prices – i.e., the "global financial cycle". These variables are, in turn, significantly influenced by U.S. monetary policy (see, e.g., Bruno and Shin, 2015; Miranda-Agrippino and Rey, 2020). Rey (2015) argues that, while most individual countries cannot control the global financial cycle, national macroprudential policies can be used to increase resilience of the domestic financial system.

Our results provide valuable insights into the trilemma versus dilemma debate and its policy implications. We find that a fixed exchange rate regime is associated with a significantly higher vulnerability to large downturns in GDP triggered by U.S. financial and monetary policy shocks. This suggests that the classical trilemma still exists and may help to explain episodes such as the 1997 Asian crisis, in which the impossible trinity of fixed exchange rates, unconstrained capital mobility, and an independent monetary policy contributed to large capital inflows followed by a sudden stop, a financial crisis, and a large decline in GDP growth (e.g., Gourinchas and Obstfeld, 2012). Our finding that downside risks to growth due to U.S. financial and monetary policy shocks are larger for countries with fixed exchange rates and high foreign currency exposure is consistent with the Asian experience. Our result that laxer limits on household debt are associated with higher downside risks to growth due to U.S. financial and monetary policy shocks mirrors the macroeconomic developments following the 2008 financial crisis, consistent with evidence on the role of household debt in the macroeconomy (e.g., Mian et al., 2017). These findings suggest that macroprudential policies limiting the extent of foreign currency exposure and the amount of household leverage could mitigate the risk of large economic downturns, consistent with structural models of leverage and macroprudential policies (e.g., Bianchi, 2011; Farhi and Werning, 2016; Korinek and Simsek, 2016).

Our paper proceeds as follows. Section 2 presents the econometric methodology, Section 3 describes the empirical specification used in the analysis, Section 4 presents the results, and Section 5 concludes.

2. Methodology

The results by Adrian et al. (2019) and others suggest that there are dynamics beyond the conditional mean which are important to understand the complex relationship between financial and macroeconomic variables. Therefore, in this section, we introduce a quantile VAR that allows us to model the conditional distribution of the endogenous variables in terms of their conditional quantiles, and to derive quantile impulse response functions. Finally, the section closes with a description of the multi-country setup used to study the international transmission of U.S. shocks to macro tail risks.

2.1. Quantile vector autoregression

Consider the following reduced-form quantile VAR of finite order p (see, e.g., Petrella and Raponi, 2019; Schüler, 2020):

$$y_t = \mu_\tau + B_{1|\tau} y_{t-1} + \cdots + B_{p|\tau} y_{t-p} + u_{t|\tau}, \tag{1}$$

where $y_t = (y_{1t}, \ldots, y_{kt})'$ is a $k \times 1$ vector of time series observed at $t = 1, \ldots, T$, $\mu_\tau = (\mu_{\tau_1}, \ldots, \mu_{\tau_k})'$ is an $k \times 1$ vector of intercepts, $B_{l|\tau}$ are $k \times k$ coefficient matrices for $l = 1, \ldots, p$, and $u_{t|\tau} = (u_{1t|\tau}, \ldots, u_{kt|\tau_k})'$ denotes a $k \times 1$ vector of innovations. Furthermore, $\tau = (\tau_1, \ldots, \tau_k)'$ refers to a $k \times 1$ vector of fixed quantile values, allowing the quantiles to potentially differ across the equations (e.g., τ_1 could have a different value than τ_2). Within the context of the quantile VAR it is assumed that $Q_\tau(u_{t|\tau}|y_{t-1}, \ldots, y_{t-p}) = (0, \ldots, 0)'$, whereby $Q_\tau(.|.)$ is defined as an elementwise application of the conditional quantile operator to the stacked elements, that is, $Q_{\tau_1}(u_{1t|\tau_1}|y_{t-1}, \ldots, y_{t-p}) = 0$, $Q_{\tau_2}(u_{2t|\tau_2}|y_{t-1}, \ldots, y_{t-p}) = 0$, and so on. This means that the quantile VAR describes the marginal conditional quantile for each element in a vector of time series variables.

To simplify notation, the system of equations can be written in companion form:

$$\mathbf{Y}_t = \boldsymbol{\mu}_\tau + \mathbf{B}_\tau \mathbf{Y}_{t-1} + \mathbf{U}_{t|\tau}, \tag{2}$$

where we define the respective matrices as:

$$\underset{(kp\times 1)}{\mathbf{Y}_t = \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}}, \underset{(kp\times 1)}{\boldsymbol{\mu}_\tau = \begin{bmatrix} \mu_\tau \\ 0 \\ \vdots \\ 0 \end{bmatrix}}, \underset{(kp\times kp)}{\mathbf{B}_\tau = \begin{bmatrix} B_{1|\tau} & B_{2|\tau} & \dots & B_{p-1|\tau} & B_{p|\tau} \\ I_k & 0 & \dots & 0 & 0 \\ 0 & I_k & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & I_k & 0 \end{bmatrix}}, \underset{(kp\times 1)}{\mathbf{U}_{t|\tau} = \begin{bmatrix} u_{t|\tau} \\ 0 \\ \vdots \\ 0 \end{bmatrix}}.$$

Using the assumption on the innovations as well as the elementwise application of the conditional quantile operator, the conditional quantile of $\mathbf{Y}_t$ given $\mathbf{Y}_{t-1}$ is:

$$Q_\tau \left[\mathbf{Y}_t|\mathbf{Y}_{t-1}\right] = \boldsymbol{\mu}_\tau + \mathbf{B}_\tau \mathbf{Y}_{t-1}. \tag{3}$$

More specifically, we define our quantile operator, for instance, on $\mathbf{Y_t}$, as

$$Q_\tau\left[\mathbf{Y}_t|\mathbf{Y}_{t-1}\right] = \begin{bmatrix} Q_\tau(y_t|\mathbf{Y}_{t-1}) \\ Q_\tau(y_{t-1}|\mathbf{Y}_{t-1}) \\ \vdots \\ Q_\tau(y_{t-p+1}|\mathbf{Y}_{t-1}) \end{bmatrix}, \tag{4}$$

where $Q_\tau(y_t|\mathbf{Y}_{t-1}) = (Q_{\tau_1}(y_{1t}|\mathbf{Y}_{t-1}), Q_{\tau_2}(y_{2t}|\mathbf{Y}_{t-1}), \dots, Q_{\tau_k}(y_{kt}|\mathbf{Y}_{t-1}))'$, $Q_\tau(y_{t-1}|\mathbf{Y}_{t-1}) = (Q_{\tau_1}(y_{1,t-1}|\mathbf{Y}_{t-1}), Q_{\tau_2}(y_{2,t-1}|\mathbf{Y}_{t-1}), \dots, Q_{\tau_k}(y_{k,t-1}|\mathbf{Y}_{t-1}))'$, and so on.

We estimate the reduced-form quantile VAR using Bayesian methods. Particularly, we estimate the reduced-form parameters for fixed quantile values τ by drawing from a multivariate Laplace distribution using a Metropolis-within-Gibbs sampler with uninformative priors.[3] Further details on the estimation procedure are provided in Appendix A.1.

2.1.1. Quantile impulse response function

Let us consider a thought experiment, in which the conditional quantile of y_{it} is shocked by an innovation at time t (e.g., $\mathbf{U}^\star = (u^*, 0, \dots, 0)'$), compared to leaving it unchanged. The quantile impulse response function (QIRF) for any horizon $h \geq 0$, n denoted as QIRF$(h, \mathbf{U}^\star, \tau)$, traces how this shock affects the entire system of variables $\mathbf{Y}_t$, including their quantile path-dependent effects on $\mathbf{Y}_{t+h}$, assuming no other shocks occur at any $h \geq 0$ (see also Gallant et al., 1993; Koop et al., 1996).

Specifically, let $\boldsymbol{\Xi}^\star_\tau(h)$ and $\boldsymbol{\Xi}_\tau(h)$ denote the quantile path of Y_{t+h} with and without the shock, respectively:

$$\boldsymbol{\Xi}^\star_\tau(h) = \begin{cases} \boldsymbol{\mu_\tau} + \mathbf{B}_\tau \mathbf{Y}_{t-1} + \mathbf{U}^\star & \text{if } h = 0, \\ \boldsymbol{\mu_\tau} + \mathbf{B}_\tau \boldsymbol{\Xi}^\star_\tau(h-1) & \text{if } h > 0, \end{cases} \qquad \boldsymbol{\Xi}_\tau(h) = \begin{cases} \boldsymbol{\mu_\tau} + \mathbf{B}_\tau \mathbf{Y}_{t-1} & \text{if } h = 0, \\ \boldsymbol{\mu_\tau} + \mathbf{B}_\tau \boldsymbol{\Xi}_\tau(h-1) & \text{if } h > 0. \end{cases}$$

In line with Chavleishvili and Manganelli (2024), we define the QIRF as:

$$\text{QIRF}(h, \mathbf{U}^\star, \tau) \equiv \boldsymbol{\Xi}^\star_\tau(h) - \boldsymbol{\Xi}_\tau(h). \tag{5}$$

For $h = 0$, this simplifies to:

$$\text{QIRF}(0, \mathbf{U}^\star, \tau) \equiv \boldsymbol{\Xi}^\star_\tau(0) - \boldsymbol{\Xi}_\tau(0) = \mathbf{U}^\star. \tag{6}$$

More generally, for $h \geq 0$, this yields:

$$\text{QIRF}(h, \mathbf{U}^\star, \tau) = \mathbf{B}^h_\tau \mathbf{U}^\star. \tag{7}$$

Notice that in contrast to the law of iterated expectations, there is no corresponding law of iterated quantiles. Instead, the definition of our QIRF, in line with Chavleishvili and Manganelli (2024), is motivated by the thought experiment described above. The QIRF differs from the impulse response function (IRF) in a standard VAR if and only if the matrix $\mathbf{B}_\tau$ of the quantile VAR differs from the coefficient matrix, say $\mathbf{B}$, of the corresponding VAR. If this is the case, the dynamics of the conditional distribution of $\mathbf{Y}_t$ are such that they cannot be fully captured by shifts in the conditional mean of an otherwise time-invariant distribution. Hence, only in this case does a standard VAR not provide information on the dynamics of the entire distribution of $\mathbf{Y}_t$ (as opposed to its conditional mean), and there is a role for studying tail risks. As suggested by Adrian et al. (2019) and others, this situation is relevant in particular for the link between financial and macroeconomic variables.

The QIRF allows for estimating the effects on different conditional quantiles. Specifically, let $\tau_{t:t+h} = (\tau_t, \dots, \tau_{t+h})$ $(k \times h+1)$, where τ_t is a $k \times 1$ vector of quantiles chosen for period t. Then, $\tau_{t:t+h}$ defines a path of quantiles for each variable contained in the vector $\mathbf{Y}_t$. The QIRF then becomes:

$$\text{QIRF}(h, \mathbf{U}^\star, \tau_{t:t+h}) = \mathbf{B}_{\tau_{t+h}} \mathbf{B}_{\tau_{t+h-1}} \dots \mathbf{B}_{\tau_{t+1}} \mathbf{U}^*. \tag{8}$$

In Appendix A.2, we provide a two-dimensional example of a quantile VAR and its QIRF.

2.1.2. Orthogonalization of quantile VAR innovations

In the context of conventional VARs, there exist mappings between the reduced form shocks and the structural shocks, say u_t and ε_t. For instance, such a mapping can be obtained by applying the Cholesky decomposition to the covariance matrix of the reduced-form residuals, Σ_u.

To perform structural analysis based on the quantile VAR model, we transform the error terms using a related approach. Following Schüler (2020), we use a co-exceedance matrix in the spirit of Blomqvist (1950) and Koenker and Bassett (1990), which captures the

[3] The priors are $\beta \sim \mathcal{N}(0, I_{k(kp+1)} \cdot 10)$ and $\Sigma \sim \mathcal{IW}(k, I_k)$, with β being the column vector $\text{vec}((\mu_\tau, B_{1|\tau}, \dots, B_{p|\tau})')$. The Metropolis-within-Gibbs sampler takes 10,000 draws and discards the first 5000 draws as burn-in draws. We check trace plots and the quantile conditions on the error terms to assure the convergence of the sampler.

co-variation of residuals from the quantile VAR around quantiles, just as Σ_u summarizes the common fluctuations of residuals around the conditional mean in conventional VARs. The co-exceedance matrix is given by:

$$\Omega_\tau = (\omega_{ij}) \equiv \frac{\mathrm{E}[\psi_{\tau_i}(u_{it|\tau_i})\psi_{\tau_j}(u_{jt|\tau_j})]}{f_{u_{it|\tau_i}}(0) f_{u_{jt|\tau_j}}(0)}, \tag{9}$$

where $\psi_{\tau_i}(u_{it|\tau_i}) \equiv \tau_i - \mathbb{1}(u_{it|\tau_i} < 0)$, with $\mathbb{1}$ being the indicator function, and $i, j \in \{1, \dots, k\}$. Furthermore, $f_{u_{it|\tau_i}}(0)$ denotes the probability density function of $u_{it|\tau_i}$ evaluated at 0. The product of $\psi_{\tau_i}(u_{it|\tau_i})$ and $\psi_{\tau_j}(u_{jt|\tau_j})$ measures the strength with which two error terms jointly exceed their respective quantiles contemporaneously.

By applying the Cholesky decomposition to the co-exceedance matrix, we obtain orthogonal shocks $\varepsilon_{t|\tau} = (\varepsilon_{1t|\tau}, \dots, \varepsilon_{kt|\tau})'$, where each $\varepsilon_{it|\tau}$ depends on the full vector of quantiles τ. Specifically, the Cholesky decomposition $\Omega_\tau = P_\tau P_\tau'$ yields

$$\varepsilon_{t|\tau} = P_\tau^{-1} \widetilde{\psi}_\tau(u_{t|\tau}), \tag{10}$$

where $\widetilde{\psi}_\tau(u_t) = (\psi_{\tau_1}(u_{1t|\tau_1})/f_{u_{1t|\tau_1}}(0), \dots, \psi_{\tau_k}(u_{kt|\tau_k})/f_{u_{kt|\tau_k}}(0))'$. This relation means that in our QIRF framework, for instance, when assuming $\varepsilon_{1t|\tau} = \delta$, we set the u^* $(k \times 1)$ in $\mathbf{U}^*$ to

$$u^* = P_\tau \begin{pmatrix} \delta \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{11}$$

All of this implies that the structural shocks $\varepsilon_{t|\tau}$ in our framework are not linearly related to the reduced-form shocks $u_{t|\tau}$; instead, the reduced-form shocks are transformed via the ψ-operator to capture the quantile dependencies that are essential for our quantile impulse response analysis. This approach can be interpreted as a form of quantile scenario analysis, where the vector τ allows us to identify and analyze co-exceedances across quantiles. We provide a two-dimensional example of the orthogonalization in Appendix A.3.

2.1.3. Relation to other approaches

Our approach to derive quantile impulse responses is based on Schüler (2020). An alternative approach is proposed by Chavleishvili and Manganelli (2024). They develop a recursive quantile VAR model in which the contemporaneous relations between the endogenous variables in y_t are estimated explicitly. We show below that the approach used in this study and the methodology proposed by Chavleishvili and Manganelli (2024) give the same inference, meaning the results remain overall the same even when the alternative methodology is applied.

Unlike Chavleishvili and Manganelli (2024), our approach does not rely on a recursive structure for y_t. This is because we do not estimate the multiple time series system on an equation-by-equation basis, as Chavleishvili and Manganelli (2024) do. The equation-by-equation estimation used in their methodology necessitates the recursive structure to recover the joint distribution of y_t. Without this recursive structure, it would not be possible to reconstruct the joint distribution in their framework (see, for instance, the discussion on Rosenblatt's theorem in Wei, 2008). Thus, the reliance on a recursive structure is a defining feature of their approach.

In contrast, our approach employs the multivariate Laplace distribution to model the joint distribution of y_t. This allows us to estimate a reduced-form VAR while still being able to infer the joint distribution of y_t. The use of the multivariate Laplace distribution provides a highly flexible framework for capturing the dynamics in a quantile VAR model (see Appendix A.1 for details).

Another approach to compute quantile impulse responses has been proposed by Loria et al. (2025). They suggest a two-step approach which consists of (i) estimating the quantile of industrial production growth conditional on financial conditions using quantile regressions, and (ii) using linear local projections to obtain impulse responses of the conditional quantile. In our multi-country setup, this two-step approach would require measuring local financial conditions for all countries in the sample. The necessary data for doing so is however only available for a limited set of countries for a sufficiently long period. For instance, Adrian et al. (2022) compute quantiles of GDP growth conditional on local financial conditions for 11 advanced economies for a sample starting in the early 1980s. The quantile VAR approach is able to accommodate more countries, in our case 44 advanced and emerging countries, in a more flexible way compared to the two-step local projection approach. Furthermore, the approach by Loria et al. (2025) does not allow to identify quantile shocks. Nevertheless, we show below that results are robust to the use of a one-step local projection approach, in which the conditional quantiles of h-step ahead GDP growth are projected directly on U.S. financial and monetary policy shocks.

2.2. Multi-country analysis

Our objective is to study cross-country differences in the transmission of U.S. financial and monetary policy shocks to macroeconomic tail risks, and to shed light on the country characteristics that help to explain these differences. We proceed in two stages. In the first stage, we use a multi-country approach similar to, e.g., Cesa-Bianchi et al. (2018), Degasperi et al. (2020), and Metiu (2021). For a specific vector of quantiles τ, we estimate a quantile VAR for each country and compute country-specific QIRFs. To obtain a summary measure across all countries, we compute the point-for-point median across country-specific QIRFs, which can be interpreted as the median group estimator as in Degasperi et al. (2020).[4] Around the median effects, we report one-standard-deviation bands that reflect cross-country heterogeneity. (To quantify the statistical uncertainty surrounding our estimates, in Fig. 2 we report QIRFs for each country along with conventional Bayesian probability intervals.)

In the second stage, conditional on a specific vector of quantiles, we investigate the sources of heterogeneity in QIRFs across countries using a regression-based approach as in, e.g., Dedola and Lippi (2005) and Dedola et al. (2017). In line with previous studies, we use a four-quarter sum of the QIRFs as a summary measure of the strength of the estimated effects for each country.[5] We regress this measure on various country characteristics to explain cross-country variation in the QIRFs for different quantiles of interest.

It is important to note that the approach taken in our second stage, does not explicitly account for estimation uncertainty in the QIRFs, which may result in underestimated standard errors in the second stage. Nonetheless, we believe this analysis still offers meaningful insights into the sources of cross-country heterogeneity.

3. Empirical specification

We analyze quarterly time series data for 44 advanced and emerging economies for the period between 1980:Q1 and 2018:Q4.[6] The baseline quantile VAR for country n includes four variables in the following order: the EBP constructed by Gilchrist and Zakrajsek (2012) and updated by Favara et al. (2016) as a measure of U.S. financial conditions; the log-difference of real GDP of country n; the log-difference of the consumer price index (CPI) of country n; and the short-term interest rate of country n (where $n = 1, \dots, 44$). For each country, we collect data for real GDP and CPI from the OECD and the IMF.[7] Interest rate data come from Eurostat, OECD, IMF, and Emter et al. (2019).

Shock identification. We use this baseline specification to estimate the effects of a U.S. financial shock on macroeconomic tail risk abroad. Following earlier studies, we use the EBP as a measure of U.S. financial conditions (e.g., Metiu et al., 2016; Born and Enders, 2019; Loria et al., 2025). The EBP captures "variation in the average price of bearing exposure to U.S. corporate credit risk, above and beyond the compensation for expected defaults" (Gilchrist and Zakrajsek, 2012, p. 1700). A high EBP marks periods of tight financial conditions arising predominantly due to crises of domestic origin, such as the savings and loan crisis of the 1980s and early 1990s, the burst of the dotcom bubble in the early 2000s, and the 2007–2008 financial crisis (see Fig. B.1 in the Appendix B). Hence, the EBP can be seen as a measure of U.S. financial conditions.[8]

In line with earlier studies using conventional VARs, we identify a U.S. financial shock by ordering the EBP before the country-specific variables in our quantile VAR. The identifying assumption entails that the EBP may contemporaneously affect macroeconomic outcomes in individual countries, while developments outside of the United States may affect the EBP only with a lag of one quarter (e.g., Rey, 2015; Cesa-Bianchi et al., 2018; Loria et al., 2025). This strategy rests on the premise that an innovation to the EBP can be treated as an exogenous shock from an individual country's perspective.[9] However, it does not require us to take a stand on the underlying structural sources of innovations to the EBP, as pointed out by Cesa-Bianchi et al. (2018).[10] We order the EBP below the country-specific variables in a robustness check.

We estimate the effects of U.S. monetary policy shocks by adding to each country-specific quantile VAR a measure of U.S. monetary policy shocks derived from a high-frequency instrument that accounts for informational rigidities (Miranda-Agrippino and Ricco, 2021).[11] Since the shock series derived from Miranda-Agrippino and Ricco (2021) is an exogenous shock sequence, we order it before the EBP and the country-specific block, applying the same recursive identification strategy.

Parameterization of the quantile VAR. To characterize the QIRF for the lower tail of the conditional distribution of real GDP growth, say $y_{j,t}$, we consider the 10 % quantile. Specifically, we set:

$$\tau_{i=j,t} = \tau_{i=j,t+1} = \cdots = \tau_{i=j,t+h} = 0.1, \tag{12}$$

[4] Following Degasperi et al. (2020), we use the median because it is more robust to the impact of potential outliers. However, we obtain similar results when using the mean to aggregate across countries as in, e.g., Pesaran and Smith (1995), Gambacorta et al. (2014), Cesa-Bianchi et al. (2018), and Metiu (2021).

[5] Put differently, we use the cumulated quantile impulse responses of GDP growth over four quarters. This is distinct from calculating the quantile impulse responses of the four-quarter sum of GDP growth.

[6] The countries in our (unbalanced) sample are: Australia (AUS), Austria (AUT), Belgium (BEL), Canada (CAN), Switzerland (CHE), Chile (CHL), China (CHN), Colombia (COL), Czechia (CZE), Germany (DEU), Denmark (DNK), Spain (ESP), Estonia (EST), Finland (FIN), France (FRA), United Kingdom (GBR), Greece (GRC), Hong Kong (HKG), Hungary (HUN), Indonesia (IDN), Ireland (IRL), Island (ISL), Israel (ISR), Italy (ITA), Japan (JPN), South Korea (KOR), Lithuania (LTU), Luxembourg (LUX), Latvia (LVA), Mexico (MEX), Malaysia (MYS), Netherlands (NLD), Norway (NOR), New Zealand (NZL), Philippines (PHL), Poland (POL), Portugal (PRT), Russia (RUS), Singapore (SGP), Slovakia (SVK), Slovenia (SVN), Sweden (SWE), Thailand (THA), and South Africa (ZAF).

[7] If for either of those variables no seasonally adjusted series are available, we apply the U.S. Census Bureau's X-13 seasonal adjustment procedure.

[8] While in principle the EBP could to some extent also reflect changes in risk appetite of foreign investors, it is apparent that the EBP is low and hardly moves during foreign crises, such as the 1994 Mexican peso crisis or the 1997 Asian financial crisis.

[9] Across the different country-specific quantile VARs, we find that the $u_{1t|\tau_1}$ have an average correlation of 0.97, providing strong support for the exogeneity assumption. For more details, see Fig. B.2 in Appendix B.

[10] To the extent that innovations to the EBP might capture a global tightening in financial conditions, our results could also be interpreted as arising from a global rather than U.S. financial shock. Disentangling the two is complicated by the fact that global financial conditions are greatly determined by U.S. financial conditions (e.g., Bruno and Shin, 2015; Rey, 2015).

[11] We estimate a VAR for U.S. data using the baseline model specification of Miranda-Agrippino and Ricco (2021) that includes the EBP. We extract the realized monetary policy shock sequence from the VAR using an instrumental variables approach with a proxy for U.S. monetary policy shocks proposed by Miranda-Agrippino and Ricco (2021) that we obtain from the authors' website (see http://silviamirandaagrippino.com/).

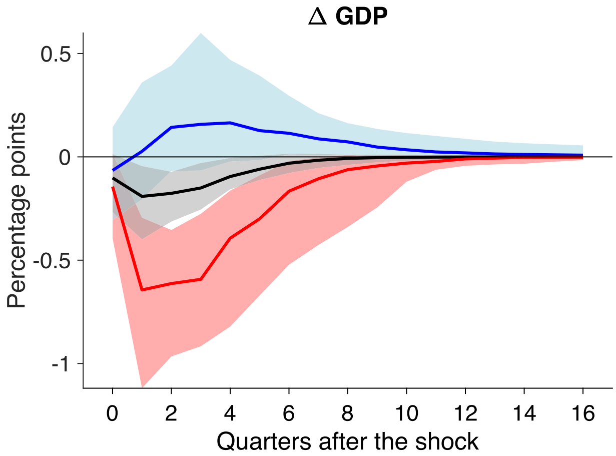


Fig. 1. Effects of a U.S. financial shock: cross-country heterogeneity. The figure depicts the cross-country median of the quantile impulse response functions for the median (black solid line), the 10 % quantile (red solid line), and the 90 % quantile (blue solid line) of the conditional GDP growth distribution. Shaded areas correspond to one standard deviation around the cross-country median, indicating heterogeneity in the estimated impulse responses across countries. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

while all other variables take on median paths:

$$\tau_{i\neq j,t} = \tau_{i\neq j,t+1} = \cdots = \tau_{i\neq j,t+h} = 0.5 \tag{13}$$

for $i = 1, \cdots, k$. To characterize the upper tail of $y_{j,t}$ we set $\tau_{i=j,\cdot} = 0.9$ instead of 0.1 in Eq. (12). These specifications of the matrix $\tau_{t:t+h}$ allow us to study the dynamics of the system at the lower (upper) tail of GDP growth, while keeping changes—relative to our median benchmark ($\tau_{i,t+h} = 0.5$ for all i and h) – to a minimum (see, e.g., Montes-Rojas, 2019; Schüler, 2020).[12]

Our approach can easily accommodate other scenarios. There are two dimensions along which alternative assumptions could be considered. First, one could change assumptions about the behavior of the co-variates of the system, for instance, by assuming upper (or lower) tail responses for these variables as well (i.e., different assumptions on $\tau_{i\neq j,t} = \tau_{i\neq j,t+1} = \cdots = \tau_{i\neq j,t+h}$, in the i-dimension). In contrast to such setups, our approach is conservative in the sense that we would expect even stronger tail responses if we assumed that, for instance, financial conditions were at their upper tail, i.e. especially tight, when GDP growth is at its lower tail (see, e.g. Schüler, 2020). Second, one could modify the number of tail realizations of GDP growth after a shock (i.e. different assumptions for $\tau_{i=j,t+h}$, in the h-dimension). Clearly, in order to study tail responses at least some tail realizations have to be considered. We consider two quarters to be the minimum of subsequent tail realizations that still allow us to study interesting propagation dynamics. In a robustness check, we therefore study QIRFs when the conditional quantile of GDP growth is at its tail for two quarters after the shock, and switches to the median thereafter. We find that our results are robust to using this conservative approach (see Section 4).

4. Results

This section describes the results, beginning with the effects of U.S. financial and monetary policy shocks on macroeconomic tail risks in foreign economies. We then analyze whether the effects are systematically related to certain country characteristics.

4.1. Effects of U.S. financial shocks on foreign output growth

Fig. 1 displays the international response of GDP growth for the median (black solid line), the 10 % quantile (red solid line), and the 90 % quantile (blue solid line) of the conditional GDP growth distribution following an unexpected tightening in U.S. financial conditions, associated with an orthogonal one-standard-deviation increase in the EBP. The solid lines reflect the cross-country median of the QIRFs of GDP growth. The shaded areas represent one-standard-deviation bands of the cross-country distribution of the QIRFs of GDP growth, indicating heterogeneity in the QIRFs across countries.

Looking at the cross-country median of the estimated impulse responses, we find that an orthogonal tightening in U.S. financial conditions leads to a reduction in GDP growth abroad for the median of the GDP growth distribution (black solid line).[13] The effect of the shock on the upper tail (90 % quantile) is positive and less pronounced than the effect on the median. By contrast, the effect on the lower tail (10 % quantile) is substantially stronger than the effect on the median. After four quarters, the effect on the lower tail

[12] As shown above, the vector of quantiles also affects the identification of structural shocks (see Eq. 10). Considering just the lower tail of conditional GDP growth (and not changing the conditional quantile of the other variables) minimizes the changes with respect to the median benchmark in this regard as well.

[13] Fig. B.3 in Appendix B depicts the impulse responses of the endogenous variables to an EBP shock obtained via Cholesky decomposition from a standard recursive VAR. The shock reduces the conditional means of GDP growth, the inflation rate and the short-term interest rate, in line with findings by, e.g., Rey (2015). In Fig. 1, we omit the QIRFs of the remaining variables because these are in line with the effects on the conditional mean.

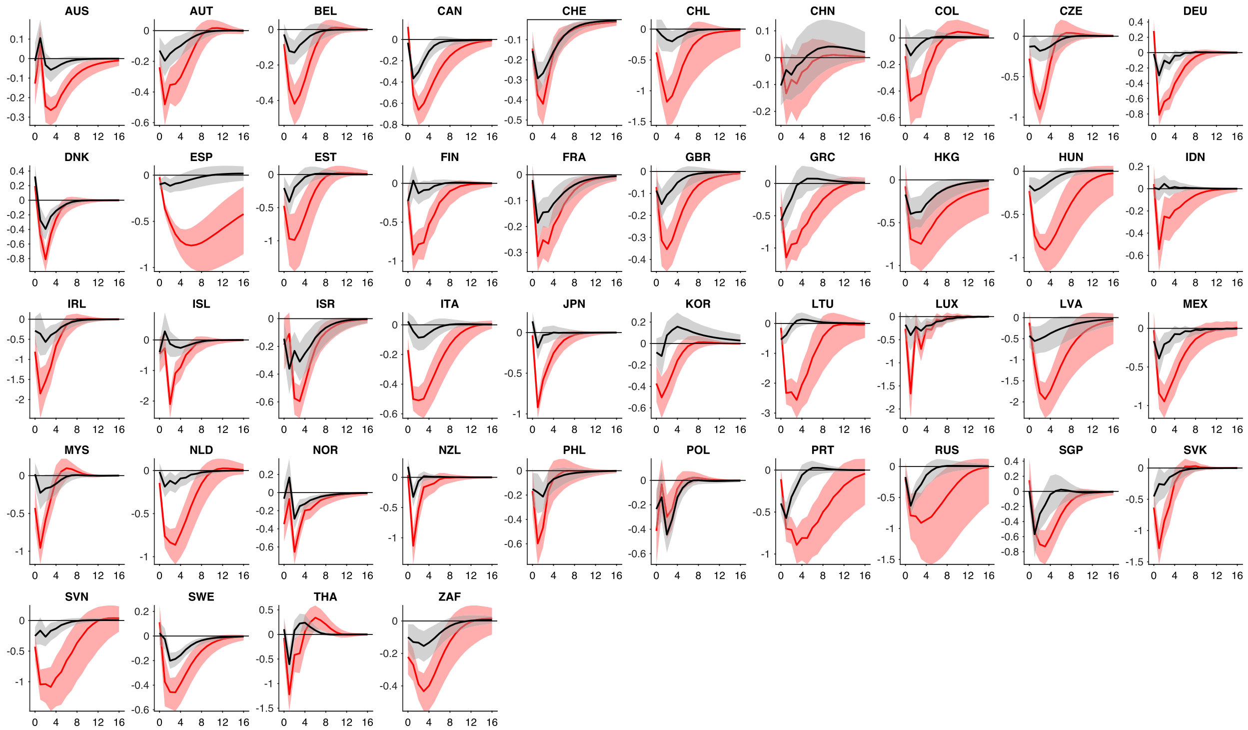


Fig. 2. Effects of a U.S. financial shock: individual country responses. Median of the posterior estimate of quantile impulse response functions (QIRFs) (solid lines) with posterior 68 % probability bands (shaded areas). The black lines represent the QIRFs for the median of the conditional GDP growth distribution, and the red lines represent the QIRFs for the 10 % quantile of the conditional GDP growth distribution. The *x*-axis shows quarters after the shock. Shocks are scaled to one standard deviation of the unconditional EBP. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

is roughly four times stronger than on the median. Thus, the conditional distribution of GDP growth changes its shape and becomes more negatively skewed as the lower tail widens in response to the shock. Moreover, the U.S. financial shock accounts for more than three times as much of the variation in GDP growth for the 10 % quantile than for the median or the 90 % quantile at horizons associated with business cycle frequencies (see Fig. B.4 in Appendix B).

The shaded areas around the cross-country median in Fig. 1 convey considerable cross-country heterogeneity. The individual QIRFs for each country (median and lower tail) are shown in Fig. 2, along with posterior 68 % probability bands. Consistent with Fig. 1, the individual QIRFs display substantial heterogeneity across countries. At the same time, the QIRFs for each country can be estimated fairly precisely. For nearly all countries and time horizons, the probability bands around the conditional 50 % and 10 % quantiles of GDP growth do not overlap and support the conclusion that the effects of a U.S. financial shock are stronger for the conditional 10 % quantile.

In some cases, the 10 % quantile impulse response is depicted above the median response, as shown in Fig. 2 – for instance, being particularly pronounced for DEU. This effect arises from our choice of a shared zero line for visualization. Unlike traditional impulse responses, QIRFs measure deviations relative to quantile-specific baselines, which differ across quantiles. By plotting these responses against a common reference point, the visualization may give the impression that lower quantile responses exceed higher quantile responses in certain periods. Importantly, this observation does not indicate quantile crossing in a statistical sense.[14]

The asymmetric effects of a shock to U.S. financial conditions on the conditional GDP growth distribution hold in a variety of robustness checks. We obtain virtually identical results when using the quantile VAR model proposed by Chavleishvili and Manganelli (2024), in which the contemporaneous relation between the endogenous variables is estimated explicitly (Fig. 3; panel (1.)).[15] The results are robust to using a balanced sample that starts in 1996:Q1 (Fig. 3; panel (2.)). Using the Chicago Fed's National Financial Conditions Index (NFCI) as a measure for U.S. financial conditions instead of the EBP leads to the same conclusions (Fig. 3; panel (3.)). Our results also hold when using PPP-GDP weighted median group estimates (Fig. 3; panel (4.)). The results remain unchanged when amending the quantile VAR with a U.S. block consisting of GDP growth, CPI inflation, and the federal funds rate, ordered above the EBP in order to remove any remaining confounding factors from the U.S. financial shock arising from other U.S. shocks (Fig. 3; panel (5.)). The results are robust to ordering the EBP last (Fig. 3; panel (6.)). The results are also not sensitive to considering the

[14] Neither our approach nor that of Chavleishvili and Manganelli (2024) explicitly prevents quantile crossing. However, given that we analyze sufficiently different quantiles, the likelihood of true quantile crossing occurring remains low.

[15] Similar to Fig. 2, we plot the individual responses using the recursive approach in Fig. B.6 in Appendix B.

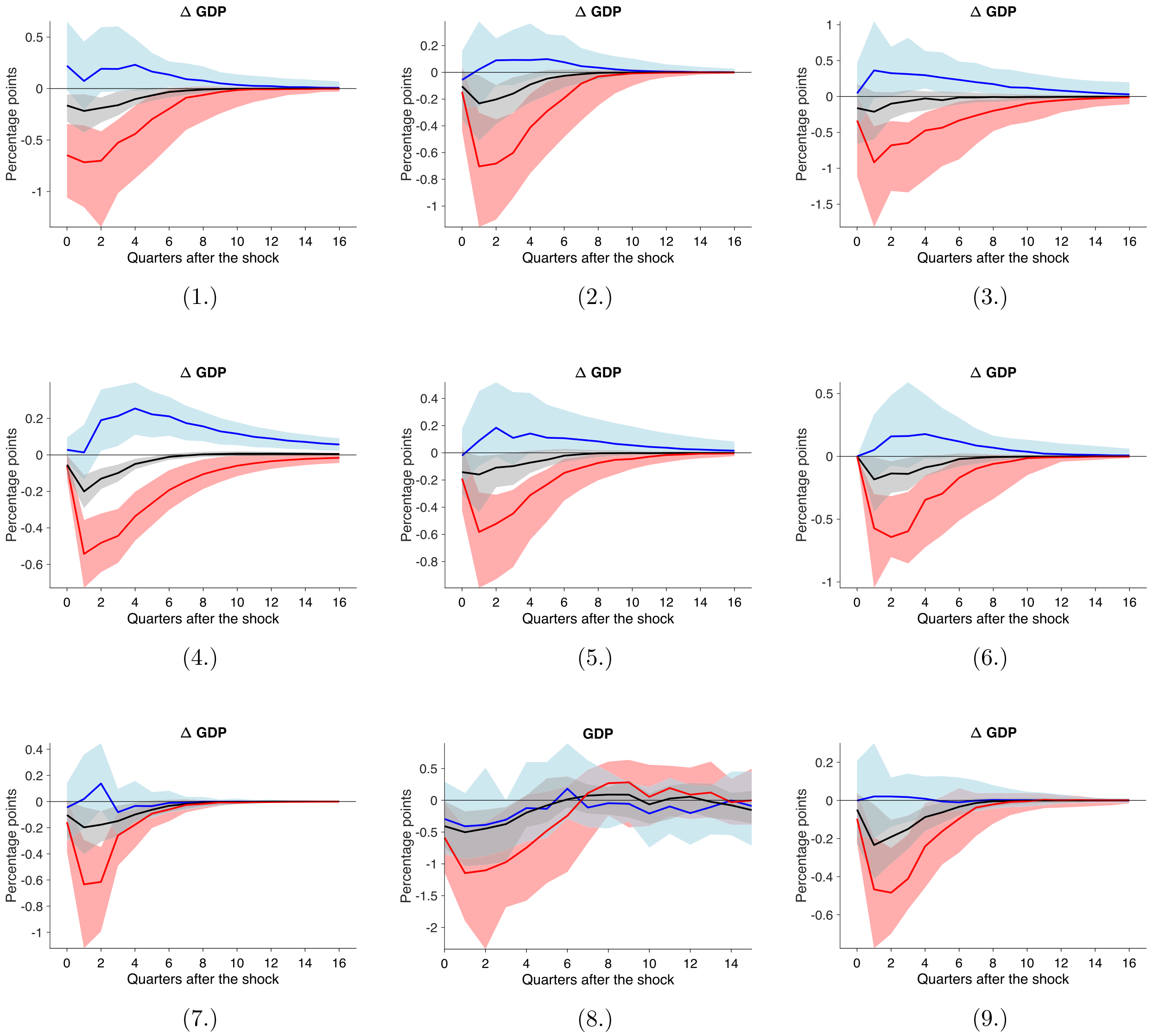


Fig. 3. Effects of a U.S. financial shock: Robustness checks. The figure depicts the cross-country median of the quantile impulse response functions for the median (black solid line), the 10 % quantile (red solid line), and the 90 % quantile (blue solid line) of the conditional GDP growth distribution. Shaded areas correspond to one standard deviation around the cross-country median, indicating heterogeneity in the estimated QIRFs across countries. Results are shown for: a recursive quantile VAR following Chavleishvili and Manganelli (2024) (Panel (1.)); a balanced sample starting in 1996:Q1 (Panel (2.)); NFCI instead of EBP (Panel (3.)); PPP-GDP weighted median group estimator (Panel (4.)); a quantile VAR augmented with U.S. GDP growth, U.S. inflation, and the U.S. short-term interest rate (Panel (5.)); the EBP ordered last (Panel (6.)); a robustness check in which, after two quarters following the shock, the quantile value for GDP growth switches from 10 % or 90 % to 50 % (Panel (7.)); the results from a quantile local projections approach (Panel (8.)); and the results from a quantile VAR including local financial conditions (Panel (9.)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

conditional 10 % (90 %) quantile of GDP growth only up to two quarters after the shock and the 50 % quantile thereafter (Fig. 3; panel (7.)). This robustness check represents a lower bound on the effects of a U.S. financial shock on macroeconomic tail risks.[16] Furthermore, when projecting the quantiles of each country's GDP growth on the EBP in quantile local projections, we find stronger effects on the lower tail than on the median or the upper tail, consistent with the estimates obtained from the quantile VAR model (Fig. 3; panel (8.)). Panel (9.) of this figure illustrates that also when accounting for local financial conditions the results remain robust.[17] Finally, panel (10.) of Fig. B.7 in Appendix B shows the impulse responses of the conditional quantiles of GDP growth to a negative U.S. technology shock, identified through the utilization-adjusted total factor productivity (TFP) series by Fernald (2014).

[16] For our sample period the average duration of a U.S. recession is around 11 months. Financial recessions typically last longer (e.g., Jorda et al., 2013), especially in emerging economies (e.g., Reinhart and Rogoff, 2014).

[17] For this exercise, we take the financial condition indices provided by the IMF which measures FCIs for 33 of the countries in our sample from 1991 until 2016.

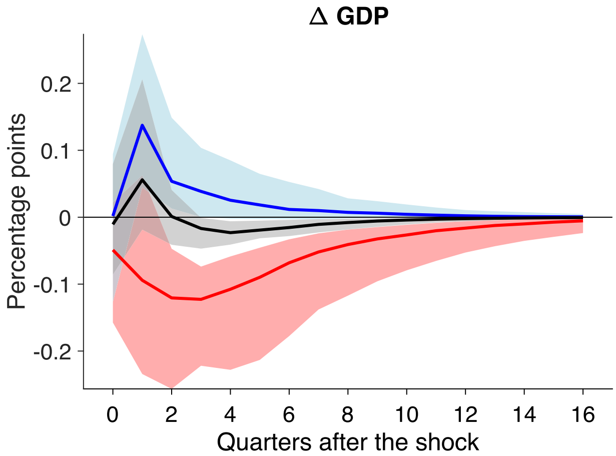


Fig. 4. Effects of a U.S. monetary policy shock: Cross-country heterogeneity. The figure depicts the point-for-point median of the quantile impulse response functions for the median (black solid line), the 10 % quantile (red solid line), and the 90 % quantile (blue solid line) of the conditional GDP growth distribution. Shaded areas correspond to one standard deviation around the point-for-point median, indicating heterogeneity in the estimated impulse responses across countries. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

This illustrates that not all shocks lead to such an asymmetric response of the conditional GDP growth distribution as we observe in response to a U.S. financial shock.

4.2. Effects of U.S. monetary policy shocks on foreign output growth

A number of recent studies have highlighted that U.S. monetary policy is an important driver of the global financial cycle (e.g., Rey, 2015; Bruno and Shin, 2015; Miranda-Agrippino and Rey, 2020). We contribute to this literature by examining the international transmission of U.S. monetary shocks on the tails of the conditional GDP growth distribution.

Fig. 4 depicts the international response of GDP growth for the median, the 10 % quantile, and the 90 % quantile of the GDP growth distribution following an unanticipated tightening in U.S. monetary policy.[18] The impact of the shock on the lower tail (10 % quantile) of conditional GDP growth is substantially stronger than the effect on the median or the upper tail (90 % quantile). Our estimates thus indicate that an unexpected monetary policy tightening in the United States leads to an increase in downside risks to growth around the world. The size of the impulse response is smaller than that for a shock to U.S. financial conditions, in line with monetary policy being only one of several factors that impact financial conditions.

Robustness checks confirm that U.S. monetary policy has stronger effects on the lower tail of the conditional GDP growth distribution. In particular, estimates yield robust conclusions upon using the recursive quantile VAR model proposed by Chavleishvili and Manganelli (2024) (Fig. 5; panel (1.)), a balanced sample starting in 1996:Q1 (Fig. 5; panel (2.)), the PPP-GDP weighted median group estimator (Fig. 5; panel (3.)), and considering the conditional 10 % (90 %) quantile of GDP growth only up to two quarters after the shock and the 50 % quantile thereafter (Fig. 5; panel (4.)).[19] We also find stronger effects on the 10 % quantile of GDP growth than on the median and the upper tail when using a quantile local projections approach (Fig. 5; panel (5.)). The estimates obtained with this approach display stronger heterogeneity across countries. This is not surprising, as local projection estimates are often less precisely estimated and more erratic (see, e.g., Ramey, 2016). Finally, also when controlling for local financial conditions results remain the same (panel (6.)).

4.3. Country characteristics and responses to U.S. financial and monetary policy shocks

Is a higher vulnerability to downside risks from U.S. financial and monetary policy shocks systematically related to certain country characteristics? To address this question, we relate the size of individual countries' GDP growth responses to several country characteristics using cross-sectional regressions in the spirit of, e.g., Dedola and Lippi (2005), Buch et al. (2014), Metiu (2016), Dedola et al. (2017), and Dahlhaus and Vasishtha (2021).

Following Dedola et al. (2017) and Cesa-Bianchi et al. (2018), we consider the following country characteristics: the degree of de facto financial openness following Lane and Milesi-Ferretti (2007); the flexibility of the exchange rate regime, measured according to the classification by Ilzetzki et al. (2019), in which countries with more freely floating currencies are assigned higher values; the level of leverage in the household sector, measured by homeownership-weighted maximum loan-to-value (LTV) ratios following

[18] Individual country IRFs can be found in Fig. B.5 in the Appendix B.

[19] Varying the variable ordering or purging the U.S. monetary policy shock series from U.S. variables would be redundant because it is an exogenous shock sequence extracted from an identified VAR model.

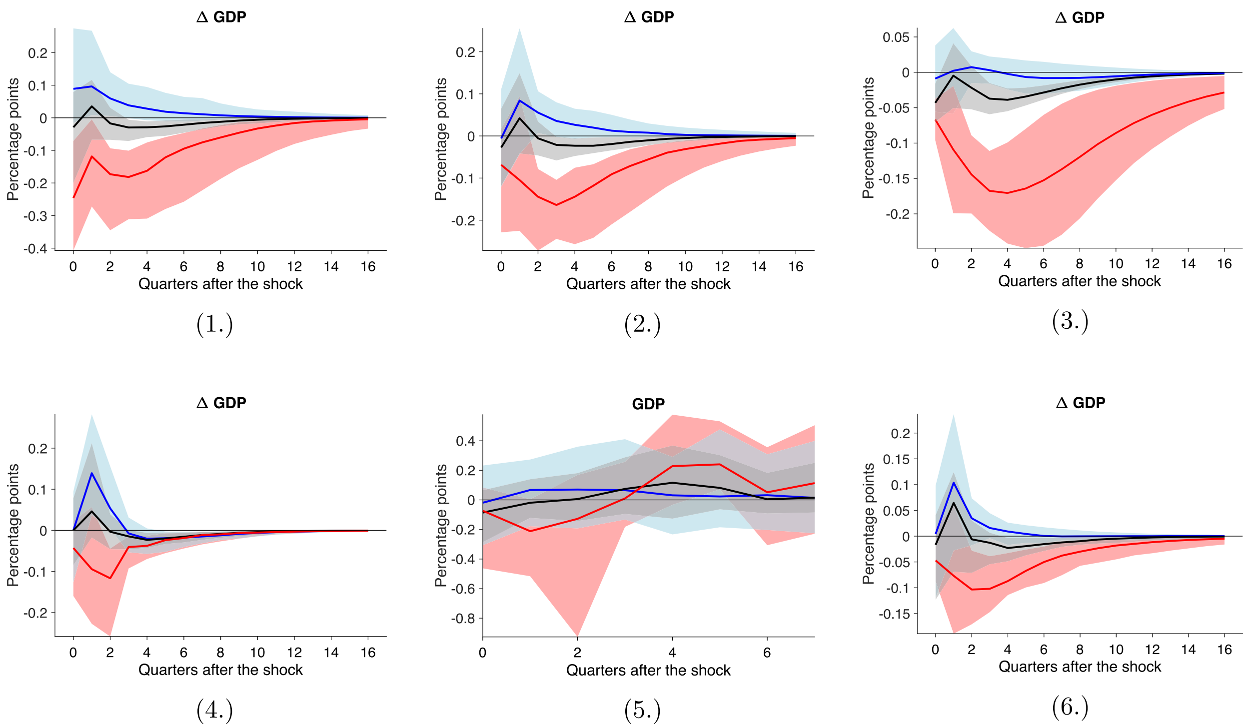


Fig. 5. Effects of a U.S. monetary policy shock: robustness checks. The figure depicts the cross-country median of the quantile impulse response functions for the median (black solid line), the 10 % quantile (red solid line), and the 90 % quantile (blue solid line) of the conditional GDP growth distribution. Shaded areas correspond to one standard deviation around the cross-country median, indicating heterogeneity in the estimated impulse responses across countries. Results are shown for: a recursive quantile VAR following Chavleishvili and Manganelli (2024) (Panel (1.)); a balanced panel starting in 1996:Q1 (Panel (2.)); the PPP-GDP weighted median group estimator (Panel (3.)); a robustness check in which, after two quarters following the shock, the quantile value for GDP growth switches from 10 % (90 %) to 50 % (Panel (4.)); results from a quantile local projections approach (Panel (5.)); and results from a quantile VAR including local financial conditions (Panel (6.)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Cesa-Bianchi et al. (2018)[20]; financial market development, measured by the IMF financial market development index; and the extent of foreign currency exposure, measured as cross-border bank claims in foreign currency over total cross-border bank claims following Cesa-Bianchi et al. (2018).[21] All country characteristics enter the regressions as time averages over the sample period between 1980:Q1 and 2018:Q4. To the extent that not all country characteristics are available over the full sample period, they enter the regressions as time averages over slightly shorter time periods. Our dependent variable is the impact of either the U.S. financial shock or the U.S. monetary policy shock on the 10 % quantile of GDP growth cumulated over the first four quarters.[22] For comparison, a second specification shows corresponding estimates on the 50 % quantile. Acknowledging that the estimated associations need not be causal, we still think it is insightful to quantify them, and to compare them to the predictions of related theoretical frameworks.

Financial shocks. Table 1 shows the ordinary least squares (OLS) estimates for the U.S. financial shock. The first column contains our main specification. We do not find a statistically significant relationship between the effects of U.S. financial shocks and country characteristics at the conditional median of GDP growth (lower panel), similar to the evidence presented by Dedola et al. (2017) and Miranda-Agrippino and Rey (2020).[23] By contrast, for the 10 % conditional quantile of GDP growth (upper panel), we find that countries with a relatively more flexible exchange rate regime exhibit a significantly more moderate (i.e., less negative) tail response of GDP growth to a U.S. financial shock. More precisely, our estimates imply that moving up 1 point on the classification between 1 and 14 of Ilzetzki et al. (2019) between fully pegged and freely falling is associated with 0.16 percentage points lower fall in the 10 % quantile of GDP growth over four quarters following a financial shock. This is consistent with a mechanism by which countries with more freely floating exchange rates face lower macroeconomic tail risks from U.S. financial shocks given that they

20 We obtain maximum LTV ratios from Alam et al. (2019) and home ownership rates from the Housing Finance Information Network.

21 Numbers on FX exposures follow Cesa-Bianchi et al. (2018) and have been cross-checked with values reported in Bénétrix et al. (2019). Numbers in Cesa-Bianchi et al. (2018) are based on “[...] cross-border bank claims in foreign currency over total cross-border bank claims.” Cross-border bank claims refer to “[...] foreign claims (all instruments, in all currencies) of all BIS reporting banks vis-à-vis all sectors.” The correlation between this FX measure and the one in Bénétrix et al. (2019) is 0.9. We use the numbers in Cesa-Bianchi et al. (2018) to maximize our country sample.

22 Fig. B.8 in the Appendix B provides a heatmap of the dependent variables used in the regressions.

23 A similar result is obtained for the 90 % quantile (which is therefore not shown).

Table 1
Country characteristics and responses to U.S. financial shocks.

Variables	QIRF of ΔGDP ($Q_{10\ \%}$)				
	(1)	(2)	(3)	(4)	(5)
Financial openness (de facto)	−0.001	−0.008	−0.001	−0.001	−0.001
	(0.006)	(0.056)	(0.006)	(0.006)	(0.006)
Exchange rate regime	0.158***	0.091**	0.159***	0.166***	0.161***
	(0.048)	(0.039)	(0.052)	(0.051)	(0.053)
Maximum LTV × home ownership	−0.032**	−0.024**	−0.032**	−0.035**	−0.034**
	(0.015)	(0.010)	(0.015)	(0.015)	(0.016)
Financial market development	−0.014	−0.010	−0.014	−0.014	−0.015
	(0.017)	(0.014)	(0.017)	(0.017)	(0.017)
Foreign currency exposure	−0.054***	−0.037**	−0.054***	−0.053***	−0.054***
	(0.018)	(0.014)	(0.018)	(0.018)	(0.018)
US financial links			−0.002		0.012
			(0.020)		(0.029)
US trade links				−0.007	−0.013
				(0.013)	(0.019)
Constant	0.770	0.361	0.795	0.977	0.976
	(1.783)	(1.287)	(1.828)	(1.841)	(1.862)
Observations	44	40	44	44	44
R-squared	0.45	0.39	0.45	0.46	0.46
Variables	**QIRF of ΔGDP ($Q_{50\ \%}$)**				
	(1)	(2)	(3)	(4)	(5)
Financial openness (de facto)	−0.002	−0.071**	−0.002	−0.002	−0.002
	(0.002)	(0.028)	(0.003)	(0.003)	(0.002)
Exchange rate regime	0.030	0.003	0.029	0.036*	0.030
	(0.020)	(0.019)	(0.022)	(0.021)	(0.022)
Maximum LTV × home ownership	−0.009	−0.009*	−0.009	−0.011*	−0.010
	(0.006)	(0.005)	(0.006)	(0.006)	(0.006)
Financial market development	−0.001	0.006	−0.001	−0.001	−0.001
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Foreign currency exposure	−0.007	−0.001	−0.007	−0.006	−0.007
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
US financial links			0.001		0.013
			(0.008)		(0.012)
US trade links				−0.005	−0.011
				(0.005)	(0.008)
Constant	−0.077	−0.178	−0.087	0.069	0.067
	(0.746)	(0.629)	(0.765)	(0.765)	(0.764)
Observations	44	40	44	44	44
R-squared	0.19	0.29	0.19	0.21	0.23

Notes: Regressions of the four-quarter sum of QIRFs for the 10 % and 50 % quantiles on country characteristics. Column (1) reports our baseline estimates. Column (2) omits potential outliers by excluding the countries IRE, LTU, LUX, and LVA; column (3) includes U.S. financial links; column (4) includes U.S. trade links; column (5) both U.S. linkages. Robust standard errors are in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively. LTV denotes loan to value.

can, in principle, pursue an independent monetary policy (e.g., Obstfeld et al., 2005; Klein and Shambaugh, 2015; Bekaert and Mehl, 2019) and therefore dampen the influence of foreign financial and monetary developments on domestic financial variables (Obstfeld, 2021). Our results are more consistent with this mechanism than with potentially stabilising effects of a pegged regime that insulates the economy from large swings in the exchange rate.

A higher share of debt denominated in foreign currency is associated with a stronger (i.e., more negative) tail response. We find that 10 percentage point higher share of liabilities denominated in foreign currency is associated with 0.54 percentage points steeper fall in the 10 % quantile of GDP growth over four quarters following a financial shock. Hence, countries with significant amounts of foreign currency debt may face stronger outflows of foreign capital when U.S. financial conditions tighten. In the case of a floating exchange rate, adverse balance sheet effects in response to a currency depreciation may represent an additional channel through which countries with a high share of foreign currency debt may suffer from a U.S. financial shock (e.g., Lane and Shambaugh, 2010; Cesa-Bianchi et al., 2018).

The homeownership-weighted maximum LTV ratio – a key determinant of domestic leverage (Cesa-Bianchi et al., 2018) – is associated with a stronger (i.e., more negative) GDP response on the tail. A 10 percentage point higher ratio is associated with 0.32 percentage points steeper fall in the 10 % quantile of GDP growth over four quarters following a financial shock. This implies that economies with higher structural levels of leverage face higher downside risks from a shock to U.S. financial conditions. This finding is consistent with the debt-driven consumption channel for business cycle dynamics highlighted by Mian et al. (2017), as well as with amplification effects due to financial frictions (e.g., Guerrieri and Iacoviello, 2017).[24]

Higher financial openness is not significantly associated with increased downside risk from U.S. financial shocks. On the one hand, financially more open economies might be more prone to spillovers from external shocks given their more integrated nature with the global financial system (Eichengreen and Gupta, 2016). On the other hand, an open financial account could enhance resilience since a more diversified portfolio of creditors could insulate the economy from shocks to specific lenders (Edwards, 2004). Our results suggest that these effects, conditional on controlling for the share of foreign currency debt, are either small or offset each other. Hence, taken together, our results suggest that it is not financial openness per se but rather exposure to certain debt instruments that makes an economy more vulnerable to downside risk from U.S. financial shocks. Finally, while one may have conjectured that a more developed financial system should provide better domestic risk sharing opportunities and correlate negatively with vulnerability to foreign financial shocks, our evidence suggests that it may not effectively insulate an economy from the effects of U.S. financial shocks on macroeconomic tail risks.

Jointly, these country characteristics explain a sizable share of the cross-country variation in the GDP growth responses of the lower tail ($R^2 = 0.45$). The regression results remain intact when potential outliers are excluded, and when we control for direct links to the U.S. via trade and financial channels (columns (2)–(5)). Our results are consistent with experiences from past financial crises that were associated with an abrupt and sizable tightening in global financial conditions. For instance, Berkmen et al. (2012) show that the global financial crisis had a particularly detrimental effect on output in countries with high levels of leverage in the domestic financial system, while exchange rate flexibility helped to cushion the impact.

Monetary policy shocks. We now turn to the question of whether the vulnerability to downside risks from U.S. monetary policy shocks varies systematically across the same country characteristics. Estimates from a regression of the size of individual countries' cumulated GDP growth responses on country characteristics are shown in Table 2.

For the conditional median, the response of GDP growth does not display a statistically significant relationship with *any* of the considered country characteristics (Table 2; lower panel), consistent with studies that do not find a clear-cut relationship using standard VAR models (e.g., Dedola et al., 2017; Miranda-Agrippino and Rey, 2020). By contrast, the coefficient on the exchange rate regime variable turns statistically significant and positive when considering the lower tail of conditional GDP growth (Table 2; upper panel). More flexible exchange rates are associated with a less strong (i.e., less negative) response of the lower tail of the GDP growth distribution to a U.S. monetary policy shock. This suggests that the classical trilemma does exist, meaning that economies with flexible exchange rates are better able to use their own monetary policies to shield themselves from foreign monetary policy shocks (e.g., Obstfeld et al., 2005, 2019; Klein and Shambaugh, 2015; Bekaert and Mehl, 2019; Obstfeld, 2021). In addition, a higher exposure to foreign currency denominated debt instruments is associated with a stronger (i.e., more negative) GDP tail response to a U.S. monetary policy shock. Similar results emerge from alternative model specifications (columns (2)–(5)).[25]

Additional robustness checks. We run additional specifications to test the robustness of the results for both financial and monetary policy shocks regarding the GDP growth responses at the lower tail, which are reported in Table B.1 in Appendix B. First, we test alternative measures of financial openness. Columns (1) and (2) show that the results are robust to using a de jure measure of financial openness (Chinn and Ito, 2006) and a measure of de facto financial openness based on the ratio of cross-border banking claims to GDP. Second, the results are robust to excluding financial centers as defined by Lane and Milesi-Ferretti (2018) (column (3)). The financial openness variable is statistically significant in this specification. This might highlight the fact that financial centers have high gross foreign asset and liability positions that are often not related to the domestic sector to a substantial degree. Third, our results are not sensitive to including the share of countries' non-U.S. trade invoiced in U.S. dollars, measured according to Boz et al. (2020). Our country sample is reduced to 36 in this case, which increases the standard error for the coefficient associated with the homeownership-weighted maximum LTV ratio and renders it insignificant for the financial shock, but the size of the coefficient estimate remains unchanged (column (4)). Fourth, we obtain very similar results when using an alternative exchange rate classification for members of the euro area (column (5)).[26] Fifth, our results are robust to using the GDP responses obtained from an alternative path for future quantiles in which 10 % quantile dynamics for GDP growth are only considered up to horizon $h = 2$ and thereafter replaced by median dynamics (column (6)). Finally, when estimating the effects on macro tail risks using quantile local projections, we obtain cross-section regression results that are in line with the baseline estimates.

[24] Cesa-Bianchi et al. (2018) find a positive association between the share of foreign currency debt as well as maximum LTV ratios and the extent to which consumption in different countries responds to international credit supply shocks in conventional VARs. They do not consider the effects on GDP growth or tail effects. The latter may be amplified by financial frictions that become binding at the tails.

[25] In column (2), we reduce the sample size by removing certain countries from the sample. The coefficient on the exchange rate regime remains at a similar magnitude, but loses statistical significance. The result on foreign currency exposure, if anything, becomes somewhat stronger.

[26] Ilzetzki et al. (2019) classify the euro area as a system of fixed exchange rates. Since we are considering U.S. financial shocks, exchange rate flexibility vis-à-vis the U.S. dollar might be important for the transmission of the shock. While the euro is freely floating against the U.S. dollar, the size of the exchange rate adjustment following a U.S. financial shock might be subject to an individual country's weight in the euro area aggregate. Therefore, we re-code the exchange rate regime as follows: For the country with the highest GDP weight (in PPP terms) we set it to fully flexible, and we record reduced flexibility with decreasing weight in aggregate euro area GDP. We obtain similar results when we use the correlation with euro area growth or inflation as weights instead of GDP.

Table 2
Country characteristics and responses to U.S. monetary policy shocks.

Variables	QIRF of ΔGDP ($Q_{10\ \%}$)				
	(1)	(2)	(3)	(4)	(5)
Financial openness (de facto)	0.000	0.018	0.000	0.000	0.000
	(0.001)	(0.027)	(0.001)	(0.001)	(0.002)
Exchange rate regime	0.030**	0.027	0.029*	0.029*	0.029*
	(0.015)	(0.016)	(0.017)	(0.016)	(0.017)
Maximum LTV × home ownership	−0.004	−0.003	−0.004	−0.004	−0.004
	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Financial market development	0.002	−0.001	0.002	0.002	0.002
	(0.004)	(0.005)	(0.004)	(0.004)	(0.004)
Foreign currency exposure	−0.005*	−0.005**	−0.005*	−0.005*	−0.005*
	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)
US financial links			0.001		−0.001
			(0.005)		(0.009)
US trade links				0.001	0.002
				(0.002)	(0.003)
Constant	−0.155	−0.092	−0.179	−0.199	−0.182
	(0.441)	(0.412)	(0.426)	(0.440)	(0.425)
Observations	44	40	44	44	44
R-squared	0.27	0.17	0.27	0.28	0.28
Variables	QIRF of ΔGDP ($Q_{50\ \%}$)				
	(1)	(2)	(3)	(4)	(5)
Financial openness (de facto)	−0.002	−0.014	−0.002	−0.002	−0.002
	(0.001)	(0.017)	(0.001)	(0.001)	(0.001)
Exchange rate regime	−0.010	−0.015	−0.012	−0.009	−0.012
	(0.008)	(0.013)	(0.009)	(0.008)	(0.010)
Maximum LTV × home ownership	−0.002	−0.002	−0.002	−0.003	−0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Financial market development	−0.003	0.001	−0.003	−0.003	−0.003
	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)
Foreign currency exposure	−0.000	0.001	−0.000	−0.000	0.000
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
US financial links			0.001		0.005
			(0.004)		(0.006)
US trade links				−0.001	−0.003
				(0.002)	(0.002)
Constant	0.344	0.312	0.274	0.333	0.332
	(0.396)	(0.375)	(0.399)	(0.410)	(0.403)
Observations	44	40	44	44	44
R-squared	0.11	0.09	0.11	0.11	0.12

Notes: Regressions of the four-quarter sum of QIRFs for the 10 % and 50 % quantiles on country characteristics. Column (1) reports our baseline estimates. Column (2) omits potential outliers by excluding the countries IRE, LTU, LUX, and LVA; column (3) includes U.S. financial links; column (4) includes U.S. trade links; column (5) includes both U.S. linkages. Robust standard errors are in parentheses. ** and * denote significance at the 5 % and 10 % levels, respectively. LTV denotes loan to value.

5. Conclusion

Financial conditions and monetary policy in the United States are important determinants of macroeconomic tail risks across the globe. In particular, we show that an unexpected tightening in U.S. financial conditions leads to a reduction of the lower tail of the conditional GDP growth distribution in individual countries that is about four times larger than the reduction of the median. We obtain similar results for U.S. monetary policy shocks. These findings connect the literature on downside risks to growth with the literature on the global financial cycle.

Our results indicate that the strength of the GDP growth response systematically varies with certain country characteristics for the lower tail of the conditional GDP growth distribution but not for the median. Policymakers concerned with the possibility of large negative output growth realizations should therefore pay particular attention to policy choices that may expose their economies to elevated GDP tail risks arising from external shocks. GDP tail risks arising from U.S. financial shocks co-vary with the extent of foreign currency liability exposure and the level of household sector leverage. Macroprudential policies that limit foreign currency exposure and household indebtedness might thus help to contain downside risks to growth after U.S. financial shocks. Less flexible exchange rates are associated with higher GDP tail risks arising from U.S. financial and monetary policy shocks. This suggests that open economies may be better equipped to weather external shocks under flexible exchange rate regimes that allow national monetary and macroprudential policies to limit downside risks to growth arising from external shocks.

CRediT authorship contribution statement

Johannes Beutel: Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization. **Lorenz Emter:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Norbert Metiu:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Conceptualization. **Esteban Prieto:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Yves Schüler:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Quantile VAR

A.1. Estimation procedure

This section summarizes the estimation procedure. For further details and proofs, see Schüler (2020).

A.1.1. Multivariate Laplace distribution for multiple equation quantile regression

The Bayesian estimation of the quantile VAR exploits a multivariate Laplace distribution, as proposed by Schüler (2020). This approach extends earlier contributions that use the univariate Laplace distribution for single equation quantile regression (see, e.g., Koenker and Machado, 1999; Yu and Moyeed, 2001).

To estimate $(\mu_\tau, B_{1|\tau}, \dots, B_{p|\tau})$, we assume that

$$u_{t|\tau} \sim \mathcal{L}_k(Cm_\tau, C\Sigma_\tau C'), \tag{14}$$

where $\mathcal{L}_k$ denotes the general multivariate Laplace distribution. m_τ and the diagonal elements of Σ_τ are defined as

$$m_\tau = (m_j) = \frac{1-2\tau_j}{\tau_j(1-\tau_j)} \quad \text{and} \quad \text{diag}(\Sigma_\tau) = (\sigma^2_{jj}) = \frac{2}{\tau_j(1-\tau_j)}. \tag{15}$$

Furthermore, C is a positive definite matrix of size $(k \times k)$ defined as $\text{diag}(c_1, \dots, c_d)$.

Schüler (2020) proposes to use a Metropolis-within-Gibbs sampler, exploiting a mixture representation of the multivariate Laplace distribution, which is given by

$$u_{t|\tau} = Cm_\tau w_t + \sqrt{w_t}C\Sigma_\tau^{1/2} z_t. \tag{16}$$

The representation allows to use commonly known results for the estimation as

$$\mathbf{y}_t | \mu_\tau, B_{1|\tau}, \dots, B_{p|\tau}, \Sigma_\tau, C, w_t, \mathcal{F}_{t-1} \sim \mathcal{N}_k, \tag{17}$$

where w_t denotes a standard exponential random variable ($w_t \sim \mathcal{E}(1)$) and z_t a k-dimensional standard multivariate normal random variable ($z_t \sim \mathcal{N}_k(0, I_k)$), with I_k being an identity matrix of dimension k. Additionally, let $\Sigma_\tau^{1/2}$ represent the square root matrix Σ_τ that yields $\left(\Sigma_\tau^{1/2}\right)\left(\Sigma_\tau^{1/2}\right)' = \Sigma_\tau$.

A.1.2. Posteriors

For introducing posteriors, we use the compact form notation as given by

$$y = (I_d \otimes X)\beta_\tau + (Cm_\tau \otimes I_T)w + \left(C\Sigma_\tau^{1/2} \otimes W^{1/2}\right) z, \tag{18}$$

where $y = \text{vec}(y_1, \dots, y_T)'$ is a $(Tk \times 1)$ vector of observations, $X = (x_1', \dots, x_T')'$ is a $(T \times (kp+1))$ matrix, where $x_t = (1, y_{t-1}', \dots, y_{t-p}')$ represents a $(1 \times (kp+1))$ vector. Furthermore, β_τ denotes the column vector $\text{vec}(\mu_\tau, B_{1|\tau}, \dots, B_{p|\tau})$ of size $(k(kp+1) \times 1)$, $w = (w_1, \dots, w_T)'$ is a $(T \times 1)$ vector and $W = \text{diag}(w)$ reflects a $(T \times T)$ diagonal matrix. Therefore, $W^{1/2} = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_T})$. $z = \text{vec}(z_1, \dots, z_T)$ denotes a $(Tk \times 1)$ vector of multivariate standard normal random variables.

Conditional posteriors of β_τ and Σ_τ. We assume an independent normal-inverse-Wishart ($\mathcal{IW}$) prior:

$$\beta \sim \mathcal{N}(\underline{\beta}, \underline{V}) \quad \text{and} \quad \Sigma \sim \mathcal{IW}(\underline{\Sigma}, \underline{\nu}). \tag{19}$$

Prior times likelihood yields the standard posterior probability density functions:

$$\beta_\tau | y, \Sigma_\tau, C, w \sim \mathcal{N}(\overline{\beta}_\tau, \overline{V}_\tau) \quad \text{and} \quad \Sigma_\tau | y, \beta_\tau, C, w \sim \mathcal{IW}(\overline{\Sigma}_\tau, \overline{\nu}), \tag{20}$$

where

$$\overline{V}_\tau = [\underline{V} + ((C\Sigma_\tau C')^{-1} \otimes (X'W^{-1}X))]^{-1} \tag{21}$$

$$\overline{\beta}_\tau = \overline{V}_\tau[\underline{V}^{-1}\underline{\beta} + ((C\Sigma_\tau C')^{-1} \otimes X'W^{-1})(y - (Cm_\tau \otimes I_T)w)] \tag{22}$$

and

$$\overline{\nu} = \underline{\nu} + T \tag{23}$$

$$\overline{\Sigma}_\tau = \underline{\Sigma} + (C')^{-1}(Y - XB_\tau - w(Cm_\tau)')'W^{-1}(Y - XB_\tau - w(Cm_\tau)')(C)^{-1}, \tag{24}$$

where $B_\tau = (\mu_\tau, B_{1|\tau}, \ldots, B_{p|\tau})'$.

Conditional probability density function of the latent variable w_t. The conditional probability density of w_t is proportional to

$$f(w_t|y_t, B_\tau, \Sigma_\tau, C, \mathcal{F}_{t-1}) \propto w_t^{-k/2} \exp\left(-\frac{1}{2}\left(a_{t|\tau} w_t^{-1} + b_\tau w_t\right)\right), \tag{25}$$

with $a_{t|\tau} = (y_t - \mu_\tau - \sum_{i=1}^p B_{i|\tau} y_{t-i})'(C\Sigma_\tau C')^{-1}(y_t - \mu_\tau - \sum_{i=1}^p B_{i|\tau} y_{t-i})$ and $b_\tau = 2 + m_\tau' \Sigma_\tau^{-1} m_\tau$. This implies that w_t, conditional on the latter parameters, is proportional to a generalized inverse Gaussian with the following parameters:

$$w_t|y_t, \Sigma_\tau, C, B_\tau, \mathcal{F}_{t-1} \sim \mathcal{GIG}\left(-k/2 + 1, a_{t|\tau}, b_\tau\right). \tag{26}$$

Conditional posterior of C. We assume a noninformative prior for C, i.e. let

$$f(C) = \text{constant}. \tag{27}$$

In this case, the conditional posterior of C follows the likelihood of a $\mathcal{L}_k(Cm_\tau, \Sigma_{\tau\star})$, where $\Sigma_{\tau\star} = C\Sigma_\tau C'$. It is given by

$$f(C|y, \beta_\tau, \Sigma_\tau) \propto \prod_{t=1}^T \frac{2\exp\left((y_t - B_\tau' x_t')'\Sigma_{\tau\star}^{-1} Cm_\tau\right)}{(2\pi)^{k/2}|\Sigma_{\tau\star}|^{1/2}} \left(\frac{(y_t - B_\tau' x_t')'\Sigma_{\tau\star}^{-1}(y_t - B_\tau' x_t')}{2 + m_\tau' \Sigma_\tau^{-1} m_\tau}\right)^{(-k/2+1)}$$

$$K_{(-k/2+1)}\left(\sqrt{(2 + m_\tau'\Sigma_\tau^{-1} m_\tau)((y_t - B_\tau' x_t')'\Sigma_{\tau\star}^{-1}(y_t - B_\tau' x_t'))}\right), \tag{28}$$

where $K_{(-k/2+1)}(\cdot)$ reflects the modified Bessel function of the second kind of order $-k/2+1$.

A.1.3. Metropolis-within-Gibbs sampler

We use a Gibbs sampler to draw β_τ and w_t. The draws of the correlations contained in Σ_τ and the scaling factors in C are as follows.

For Σ_τ, we use the conditional posterior of Σ_τ and standardize each draw, such that quantile restrictions on the Laplace distribution remain fixed. Specifically, Σ_τ may be decomposed as

$$\Sigma_\tau = S_\tau R S_\tau, \tag{29}$$

where R denotes the correlation matrix with ones on the diagonal and ρ_{lk} as off-diagonal elements and $S_\tau = \text{diag}(\sigma_{\tau_1}, \ldots, \sigma_{\tau_k})$. Therefore, each draw of Σ_τ can be rearranged using

$$R = S_\tau^{-1}\Sigma_\tau S_\tau^{-1}, \tag{30}$$

meaning that the diagonal elements of Σ_τ remain unchanged. Given a new correlation matrix R, the covariance matrix Σ_τ can be updated using Eq. (29).

For the draw of C, we use a random walk Metropolis–Hastings (MH) algorithm, since C appears in the mean and the variance of the conditional distribution of y_t (e.g., Eq. 17). Given a new draw of C, called C^*, and the last draw $C^{(n-1)}$, where $n \in \{1, \ldots, N\}$, the acceptance probability is (calibrated to be between 0.2 and 0.5)

$$\alpha_{\text{MH},C}(C^{(n-1)}, C^*) = \min\left[\frac{f\left(C^*|y, \beta_\tau^{(n)}, \Sigma_\tau^{(n)}, w^{(n)}\right)}{f\left(C^{(n-1)}|y, \beta_\tau^{(n)}, \Sigma_\tau^{(n)}, w^{(n)}\right)}, 1\right]. \tag{31}$$

Next, we depict the algorithm, which we specify to only accept stationary draws of β_τ.

A.2. Quantile impulse response function in the bivariate case

For illustration, assume $y_t = (y_{1t}, y_{2t})$, $u_{t|\tau} = (u_{1t|\tau_1}, u_{2t|\tau_2})$ and $p = 1$. The quantile VAR can then be written as

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} u_{1t|\tau_1} \\ u_{2t|\tau_2} \end{pmatrix}. \tag{32}$$

Applying our conditional quantile operator $Q_\tau\left(y_t|y_{t-1}\right)$ to the system yields

$$\begin{pmatrix} Q_{\tau_1}(y_{1t}|Y_{t-1}) \\ Q_{\tau_2}(y_{2t}|Y_{t-1}) \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} \tag{33}$$

$$+ \begin{pmatrix} Q_{\tau_1}(u_{1t|\tau_1}|Y_{t-1}) \\ Q_{\tau_2}(u_{2t|\tau_2}|Y_{t-1}) \end{pmatrix}, \tag{34}$$

Algorithm 1: Metropolis-within-Gibbs sampler for Bayesian quantile VAR

A. Define prior distribution for β_τ and Σ_τ and set starting values $\beta_\tau^0, \Sigma_\tau^0$ and C^0. Set variance of the random walk innovation used in the MH step, c.

B. Repeat for $n = 1, 2, \dots, N$.
1. Gibbs Step 1: For $t = 1, \dots, T$: Draw $w_t^{(n)} | y_t, \beta_\tau^{(n-1)}, \Sigma_\tau^{(n-1)}, C^{(n-1)}$.
2. Gibbs Step 2: Draw $\beta_\tau{}^{(n)} | y, \Sigma_\tau^{(n-1)}, C^{(n-1)}, w^{(n)}$.
3. Gibbs Step 3: (i) Draw $\Sigma_\tau^{(n)} | y, \beta_\tau^{(n)}, C^{(n-1)}, w^{(n)}$; (ii) Calculate $R^{(n)} = S_\tau^{-1} \Sigma_\tau^{(n)} S_\tau^{-1}$; (iii) Set $\Sigma_\tau^{(n)} = S_\tau R^{(n)} S_\tau$.
4. MH Step 1: (i) Draw $v_{**} \sim \mathcal{N}(0, cI_k)$; (ii) Calculate $\left(c_1^*, \dots, c_k^*\right)' = \left(c_1^{(n-1)}, \dots, c_d^{(n-1)}\right)' + v_{**}$; (iii) Evaluate $\alpha_{\mathrm{MH,C}}$; (iv) Draw $u_{**} \sim \mathcal{U}(0,1)$; (v) If $u_{**} \leq \alpha_{\mathrm{MH},C}$ set $\left(c_1^{(n)}, \dots, c_k^{(n)}\right)' = \left(c_1^*, \dots, c_k^*\right)'$; (vi) else set $\left(c_1^{(n)}, \dots, c_k^{(n)}\right)' = \left(c_1^{(n-1)}, \dots, c_k^{(n-1)}\right)'$.

which is

$$\begin{pmatrix} Q_{\tau_1}(y_{1t}|y_{t-1}) \\ Q_{\tau_2}(y_{2t}|y_{t-1}) \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix}, \tag{35}$$

as $Q_{\tau_1}(u_{1t|\tau_1}|y_{t-1}) = Q_{\tau_2}(u_{2t|\tau_2}|y_{t-1}) = 0$.

Now, let us assume that a shock hits the conditional quantile of y_{1t}, say $u_{1t,\tau_1} = \delta$, meaning that $u^* = (\delta, 0)'$. This can be written by $Q_\tau\left(y_t|y_{t-1}, u_{t|\tau} = u^\star\right)$, which is

$$\begin{pmatrix} Q_{\tau_1}(y_{1t}|y_{t-1}, u_{t|\tau} = u^\star) \\ Q_{\tau_2}(y_{2t}|y_{t-1}, u_{t|\tau} = u^\star) \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \delta \\ 0 \end{pmatrix}. \tag{36}$$

For horizon 0, the quantile impulse response function (QIRF), therefore, is

$$\mathrm{QIRF}(0, u^\star, \tau) \equiv \Xi_\tau^\star(0) - \Xi_\tau(0) \tag{37}$$

$$= \begin{pmatrix} Q_{\tau_1}(y_{1t}|y_{t-1}, u_{t|\tau} = u^\star) \\ Q_{\tau_2}(y_{2t}|y_{t-1}, u_{t|\tau} = u^\star) \end{pmatrix} - \begin{pmatrix} Q_{\tau_1}(y_{1t}|y_{t-1}) \\ Q_{\tau_2}(y_{2t}|y_{t-1}) \end{pmatrix} = \begin{pmatrix} \delta \\ 0 \end{pmatrix}. \tag{38}$$

To derive the QIRF for horizon h = 1, define

$$\begin{pmatrix} Q_{\tau_1}\left(y_{1t+1}|y_t = \Xi_\tau^\star(0)\right) \\ Q_{\tau_2}\left(y_{2t+1}|y_t = \Xi_\tau^\star(0)\right) \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} Q_{\tau_1}(y_{1t}|y_{t-1}, u_{t|\tau} = u^\star) \\ Q_{\tau_2}(y_{2t}|y_{t-1}, u_{t|\tau} = u^\star) \end{pmatrix} \tag{39}$$

and

$$\begin{pmatrix} Q_{\tau_1}\left(y_{1t+1}|y_t = \Xi_\tau(0)\right) \\ Q_{\tau_2}\left(y_{2t+1}|y_t = \Xi_\tau(0)\right) \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} Q_{\tau_1}(y_{1t}|y_{t-1}) \\ Q_{\tau_2}(y_{2t}|y_{t-1}) \end{pmatrix}. \tag{40}$$

The QIRF is then given by

$$\Xi_\tau^\star(1) - \Xi_\tau(1) = \tag{41}$$

$$\begin{pmatrix} Q_{\tau_1}\left(y_{1t+1}|y_t = \Xi_\tau^\star(0)\right) \\ Q_{\tau_2}\left(y_{2t+1}|y_t = \Xi_\tau^\star(0)\right) \end{pmatrix} - \begin{pmatrix} Q_{\tau_1}\left(y_{1t+1}|y_t = \Xi_\tau(0)\right) \\ Q_{\tau_2}\left(y_{2t+1}|y_t = \Xi_\tau(0)\right) \end{pmatrix} = \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} \delta \\ 0 \end{pmatrix} \tag{42}$$

Similarly, for the case when $h > 1$, we have

$$\begin{pmatrix} Q_{\tau_1}\left(y_{1t+h}|y_{t+h-1} = \Xi_\tau^\star(h-1)\right) \\ Q_{\tau_2}\left(y_{2t+h}|y_{t+h-1} = \Xi_\tau^\star(h-1)\right) \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} Q_{\tau_1}(y_{1t+h-1}|y_{t+h-2} = \Xi_\tau^\star(h-2)) \\ Q_{\tau_2}(y_{2t+h-1}|y_{t+h-2} = \Xi_\tau^\star(h-2)) \end{pmatrix} \tag{43}$$

and

$$\begin{pmatrix} Q_{\tau_1}\left(y_{1t+h}|y_{t+h-1} = \Xi_\tau(h-1)\right) \\ Q_{\tau_2}\left(y_{2t+h}|y_{t+h-1} = \Xi_\tau(h-1)\right) \end{pmatrix} = \begin{pmatrix} \mu_{1|\tau_1} \\ \mu_{2|\tau_2} \end{pmatrix} + \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix} \begin{pmatrix} Q_{\tau_1}(y_{1t+h-1}|y_{t+h-2} = \Xi_\tau(h-2)) \\ Q_{\tau_2}(y_{2t+h-1}|y_{t+h-2} = \Xi_\tau(h-2)) \end{pmatrix}. \tag{44}$$

The QIRF is then given by

$$\Xi_\tau^\star(h) - \Xi_\tau(h) = \begin{pmatrix} b_{11,1|\tau_1} & b_{12,1|\tau_1} \\ b_{21,1|\tau_2} & b_{22,1|\tau_2} \end{pmatrix}^h \begin{pmatrix} \delta \\ 0 \end{pmatrix}. \tag{45}$$

A.3. Orthogonalization of the quantile VAR innovations in the bivariate case

In this section, we detail the orthogonalization of the reduced-form innovations in the bivariate case. Here, the co-exceedance matrix becomes

$$\Omega_\tau \equiv \begin{pmatrix} \frac{\mathrm{E}[\psi_{\tau_1}(u_{1t|\tau_1})^2]}{f_{u_{1t|\tau_1}}(0)^2} & \frac{\mathrm{E}[\psi_{\tau_1}(u_{1t|\tau_1})\psi_{\tau_2}(u_{2t|\tau_2})]}{f_{u_{1t|\tau_1}}(0)f_{u_{2t|\tau_2}}(0)} \\ \frac{\mathrm{E}[\psi_{\tau_2}(u_{2t|\tau_2})\psi_{\tau_1}(u_{1t|\tau_1})]}{f_{u_{2t|\tau_2}}(0)f_{u_{1t|\tau_1}}(0)} & \frac{\mathrm{E}[\psi_{\tau_2}(u_{2t|\tau_2})^2}{f_{u_{2t|\tau_2}}(0)^2} \end{pmatrix}, \tag{46}$$

where $\psi_{\tau_i}(u_{it|\tau_i}) \equiv \tau_i - \mathbb{1}(u_{it|\tau_i} < 0)$, with $\mathbb{1}$ being the indicator function and $i \in \{1,2\}$. By applying the Cholesky decomposition to the co-exceedance matrix, we obtain orthogonal shocks $\varepsilon_{t|\tau} = (\varepsilon_{1t|\tau}, \varepsilon_{2t|\tau})'$, where each $\varepsilon_{it|\tau}$ depends on the full vector of quantiles $\tau = (\tau_1, \tau_2)$. Particularly, the Cholesky decomposition $\Omega_\tau = P_\tau P_\tau'$ yields

$$\begin{pmatrix} \varepsilon_{1t|\tau} \\ \varepsilon_{2t|\tau} \end{pmatrix} = P_\tau^{-1} \begin{pmatrix} \frac{\psi_{\tau_1}(u_{1t|\tau_1})}{f_{u_{1t|\tau_1}}(0)} \\ \frac{\psi_{\tau_2}(u_{2t|\tau_2})}{f_{u_{2t|\tau_2}}(0)} \end{pmatrix}. \tag{47}$$

For the two-dimensional system, the Cholesky decomposition of Ω_τ defines the matrix P_τ as:

$$P_\tau = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix},$$

where the elements are given by:

$$p_{11} = \sqrt{\frac{\mathrm{E}[\psi_{\tau_1}(u_{1t|\tau_1})^2]}{f_{u_{1t|\tau_1}}(0)^2}},$$

$$p_{21} = \frac{\frac{\mathrm{E}[\psi_{\tau_1}(u_{1t|\tau_1})\psi_{\tau_2}(u_{2t|\tau_2})]}{f_{u_{1t|\tau_1}}(0)f_{u_{2t|\tau_2}}(0)}}{\sqrt{\frac{\mathrm{E}[\psi_{\tau_1}(u_{1t|\tau_1})^2]}{f_{u_{1t|\tau_1}}(0)^2}}},$$

$$p_{22} = \sqrt{\frac{\mathrm{E}[\psi_{\tau_2}(u_{2t|\tau_2})^2]}{f_{u_{2t|\tau_2}}(0)^2} - \left(\frac{\frac{\mathrm{E}[\psi_{\tau_1}(u_{1t|\tau_1})\psi_{\tau_2}(u_{2t|\tau_2})]}{f_{u_{1t|\tau_1}}(0)f_{u_{2t|\tau_2}}(0)}}{\sqrt{\frac{\mathrm{E}[\psi_{\tau_1}(u_{1t|\tau_1})^2]}{f_{u_{1t|\tau_1}}(0)^2}}} \right)^2}.$$

Now, assuming that $\varepsilon_{1t|\tau} = \delta$ (and $\varepsilon_{2t|\tau} = 0$), we set the $u^\star$ in our QIRF framework to

$$u^\star = P_\tau \begin{pmatrix} \delta \\ 0 \end{pmatrix} = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} \delta \\ 0 \end{pmatrix} = \begin{pmatrix} p_{11}\delta \\ p_{21}\delta \end{pmatrix}. \tag{48}$$

This shows that, in this scenario, the first structural shock affects the second variable contemporaneously through p_{21}.

Appendix B. Additional figures and tables

See Figs. B.1–B.8 and Table B.1.

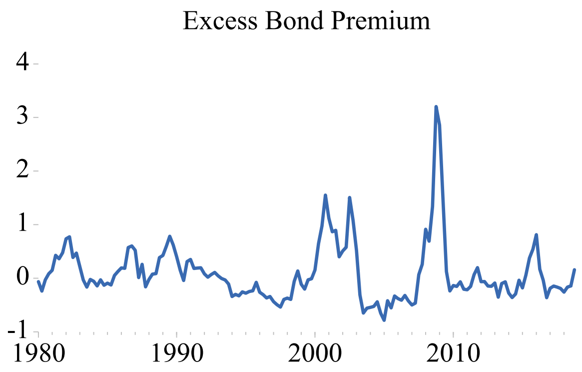


Fig. B.1. Excess bond premium. The figure depicts the Gilchrist–Zakrajsek excess bond premium for the period 1980Q1–2018Q4 (quarterly average).

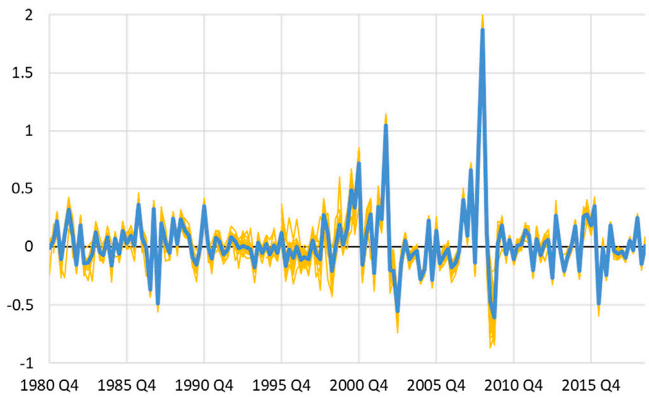


Fig. B.2. $u_{1t|\tau_1}$ across countries. The thick solid line represents the average $u_{1t|\tau_1}$ across countries, while the thin yellow lines depict the individual country $u_{1t|\tau_1}$. Given our variable ordering (with the EBP ordered first) and the Cholesky decomposition for shock identification, $u_{1t|\tau_1}$ fully reflects the EBP shock that is then transformed via the ψ-operator to $\varepsilon_{1t|\tau}$ to measure its contemporaneous impact on the other variables in the system through P_τ (see also Appendix A.3). In this sense, $u_{1t|\tau_1}$ allows us to assess the consistency of the EBP shocks across countries. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

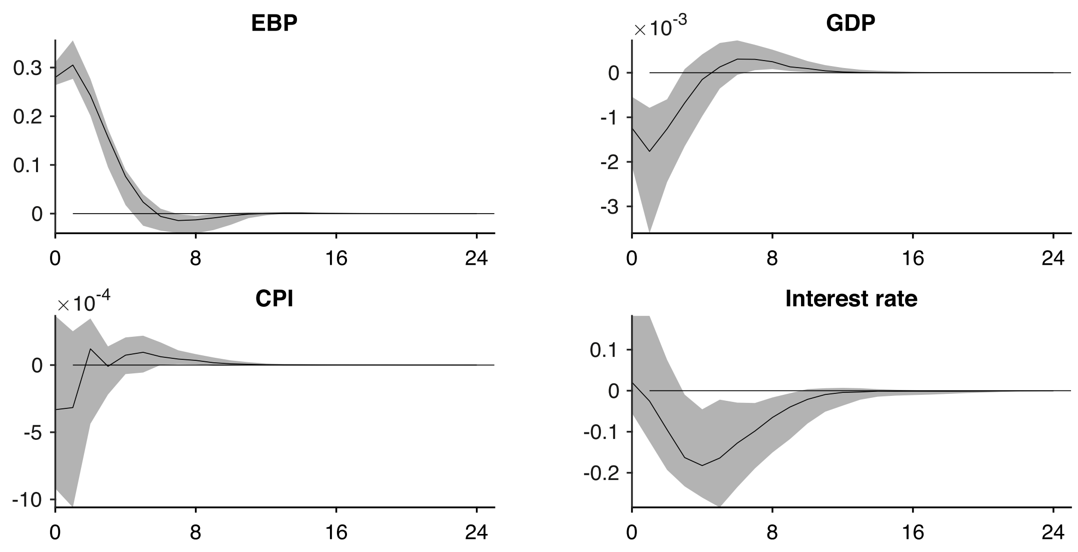


Fig. B.3. Impulse responses to an EBP shock. The solid lines are the (cross-country) median group estimates of the mean IRF obtained from a standard VAR following a tightening of U.S. financial conditions scaled to one standard deviation of the unconditional EBP. Shaded areas represent one standard deviation around the median group estimator, indicating heterogeneity in responses across countries.

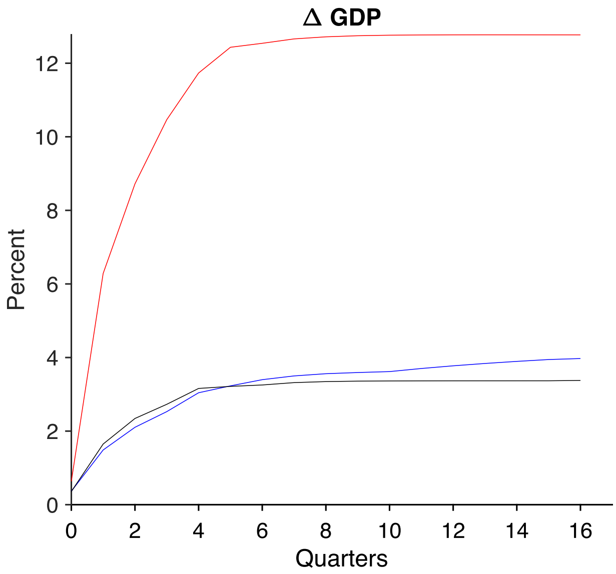


Fig. B.4. Forecast error variance share of GDP growth accounted for by U.S. financial shocks. Median group estimate of the forecast error variance share due to a one-standard-deviation increase in the EBP obtained from the baseline quantile VAR model. The black line depicts the forecast error variance share for the median of ΔGDP, the blue line shows the forecast error variance share for the 90 % quantile of ΔGDP, and the red line is the forecast error variance share for the 10 % quantile of ΔGDP. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

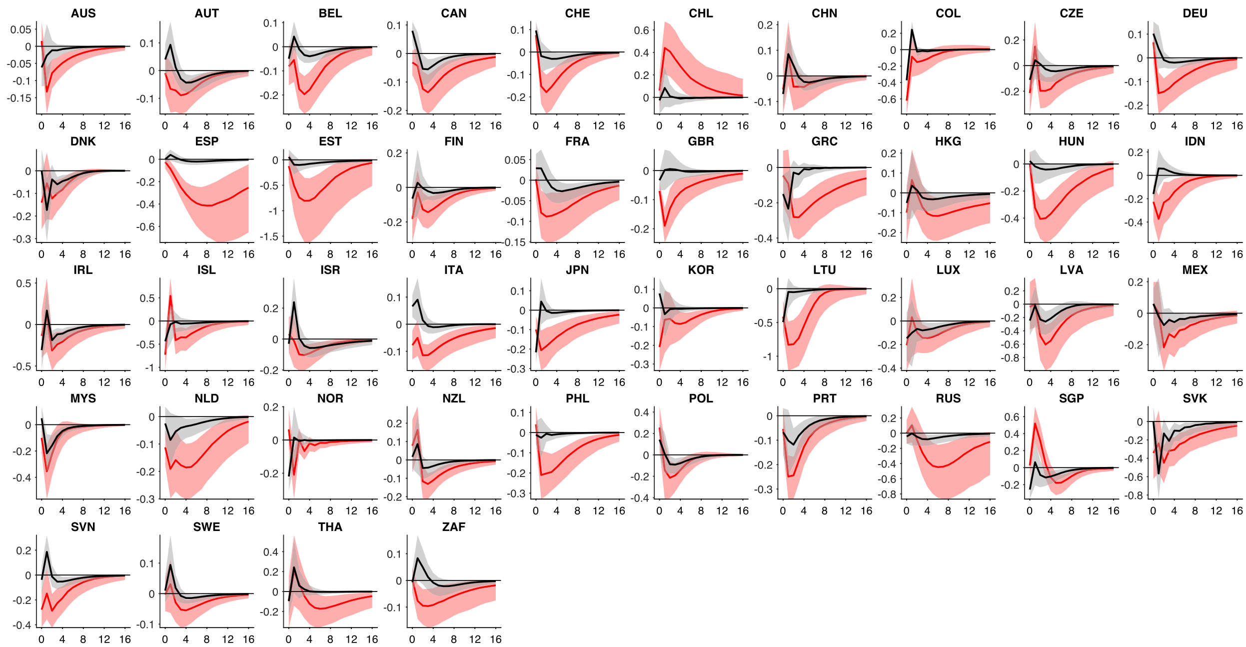


Fig. B.5. Effects of a U.S. monetary policy shock: Individual country responses. Median of the posterior estimate of quantile impulse response functions (QIRFs) (solid lines) with posterior 68 % probability bands (shaded areas). The black lines represent the QIRFs for the median of the conditional GDP growth distribution, and the red lines represent the QIRFs for the 10 % quantile of the conditional GDP growth distribution. The x-axis shows quarters after the shock. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

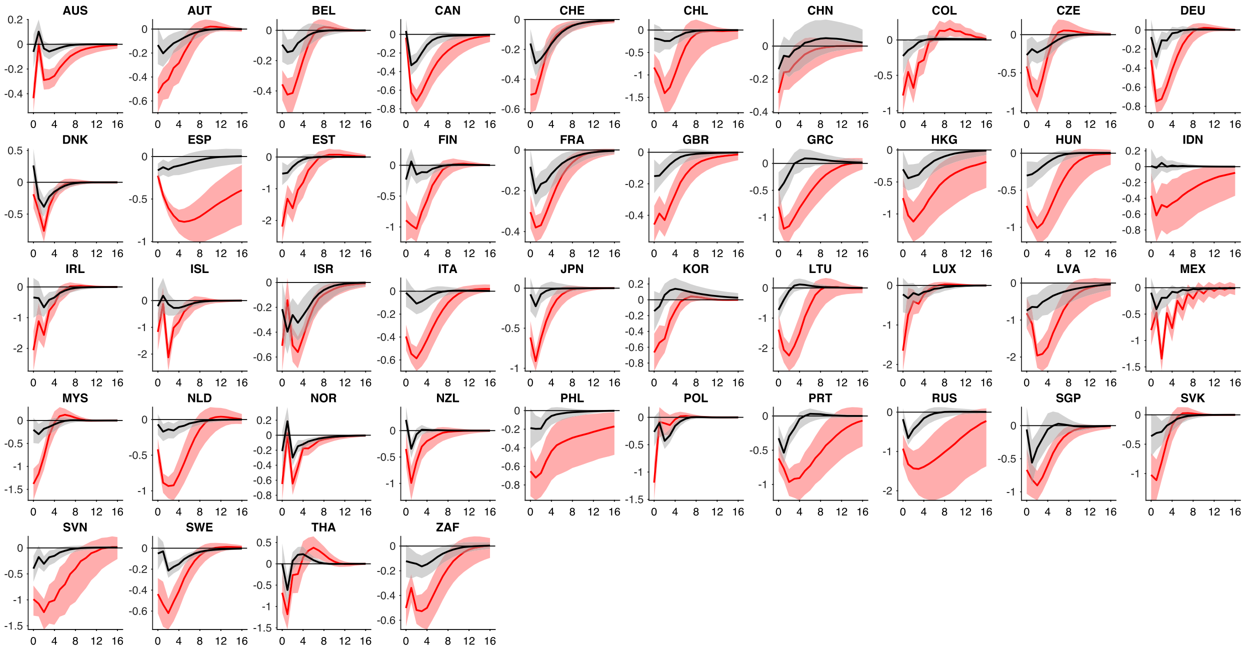
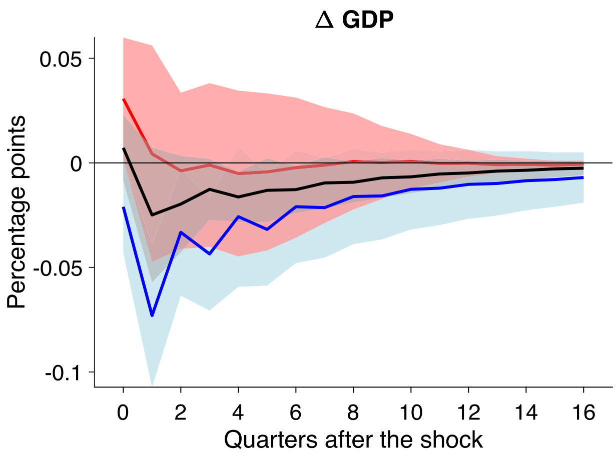


Fig. B.6. Robustness: effects of a U.S. financial shock: Individual country responses using recursive quantile VAR proposed by Chavleishvili and Manganelli (2024). Median of the posterior estimate of quantile impulse response functions (QIRFs) (solid lines) with posterior 68 % probability bands (shaded areas). The black lines represent the QIRFs for the median of the conditional GDP growth distribution, and the red lines represent the QIRFs for the 10 % quantile of the conditional GDP growth distribution. The x-axis shows quarters after the shock. Shocks are scaled to one standard deviation of the unconditional EBP. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



(10.)

Fig. B.7. Effects of a U.S. financial shock: Robustness checks. The figure depicts the cross-country median of the quantile impulse response functions for the median (black solid line), the 10 % quantile (red solid line), and the 90 % quantile (blue solid line) of the conditional GDP growth distribution. Shaded areas correspond to one standard deviation around the cross-country median, indicating heterogeneity in the estimated QIRFs across countries. Results are shown for impulse responses to a negative U.S. technology shock, obtained by using the Fernald (2014) TFP series ordered first (Panel (10.)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Country	Financial Shock: QIRF of ΔGDP (Q10%)	U.S. monetary policy shock: QIRF of ΔGDP (Q10%)
GBR	-1.0	-0.5
AUT	-1.4	-0.3
BEL	-1.2	-0.5
DNK	-1.6	-0.4
FRA	-0.9	-0.2
DEU	-1.7	-0.4
ITA	-1.7	-0.3
LUX	-2.7	-0.4
NLD	-2.6	-0.7
NOR	-1.5	-0.2
SWE	-1.2	-0.1
CHE	-1.3	-0.4
CAN	-1.7	-0.3
JPN	-2.0	-0.7
FIN	-2.7	-0.5
GRC	-3.4	-0.7
ISL	-4.0	-1.0
IRL	-5.2	-0.6
PRT	-2.4	-0.7
ESP	-1.5	-0.5
AUS	-0.6	-0.2
NZL	-1.8	0.0
ZAF	-1.3	-0.3
CHL	-3.4	1.3
COL	-1.5	-1.0
MEX	-2.6	-0.3
ISR	-1.5	-0.2
HKG	-2.1	-0.1
IDN	-1.0	-1.0
KOR	-1.5	-0.4
MYS	-2.4	-0.8
PHL	-1.4	-0.6
SGP	-1.8	0.9
THA	-2.1	0.0
RUS	-2.8	-0.2
CHN	-0.3	-0.1
CZE	-2.5	-0.5
SVK	-3.4	-1.4
EST	-3.4	-1.9
LVA	-5.0	-1.2
HUN	-2.7	-1.2
LTU	-7.3	-2.7
SVN	-3.5	-1.0
POL	-1.0	-0.3

Fig. B.8. Impact of U.S. financial and monetary policy shocks across countries. Figures refer to four-quarter sum of QIRFs for the 10 % quantile of GDP growth (in percent) used as the dependent variable in Tables 1 and 2

Table B.1
Country characteristics and responses to U.S. financial and monetary policy shocks – additional robustness checks.

Variables	QIRF of ΔGDP to U.S. financial shock ($Q_{10\ \%}$)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Financial openness	−1.328	−0.050	−0.163**	−0.003	−0.000	−0.004	0.004
	(0.839)	(0.041)	(0.076)	(0.007)	(0.005)	(0.004)	(0.003)
Exchange rate regime	0.124**	0.143**	0.101*	0.184**	0.135***	0.128***	0.102**
	(0.051)	(0.055)	(0.053)	(0.089)	(0.045)	(0.047)	(0.047)
Maximum LTV × home ownership	−0.027**	−0.029***	−0.023**	−0.024	−0.024**	−0.024***	−0.022
	(0.010)	(0.010)	(0.010)	(0.022)	(0.010)	(0.008)	(0.013)
Financial market development	0.015	0.010	0.015	0.009	0.008	0.005	0.001
	(0.014)	(0.011)	(0.012)	(0.024)	(0.013)	(0.012)	(0.011)
Foreign currency exposure	−0.021***	−0.019**	−0.036**	−0.022*	−0.013*	−0.015**	−0.020**
	(0.008)	(0.008)	(0.016)	(0.011)	(0.007)	(0.006)	(0.009)
USD trade share				−0.005			
				(0.010)			
Constant	0.231	−0.588	−1.893*	−0.844	−1.170	−0.395	−0.640
	(1.195)	(1.169)	(0.978)	(2.663)	(1.240)	(0.979)	(1.529)
Observations	44	44	37	36	44	44	43
R-squared	0.48	0.44	0.57	0.48	0.47	0.47	0.30
Variables	QIRF of ΔGDP to U.S. monetary policy shock ($Q_{10\ \%}$)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Financial openness	−0.571	0.039	−0.027	0.005**	0.004*	0.003	0.000
	(0.365)	(0.031)	(0.042)	(0.002)	(0.002)	(0.002)	(0.001)
Exchange rate regime	0.052*	0.073**	0.082**	0.080**	0.056***	0.064**	0.029*
	(0.027)	(0.030)	(0.031)	(0.036)	(0.020)	(0.026)	(0.014)
Maximum LTV × home ownership	−0.001	−0.003	0.001	0.000	−0.000	−0.002	−0.005
	(0.006)	(0.006)	(0.005)	(0.007)	(0.006)	(0.006)	(0.004)
Financial market development	0.015**	0.009	0.011*	0.001	0.011*	0.006	0.002
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.004)
Foreign currency exposure	−0.005	−0.006	−0.007*	−0.013**	−0.002	−0.006	−0.006**
	(0.005)	(0.005)	(0.004)	(0.005)	(0.004)	(0.004)	(0.002)
USD trade share				0.007			
				(0.007)			
Constant	−0.822	−0.919	−1.176*	−0.656	−1.326*	−0.695	−0.021
	(0.753)	(0.682)	(0.622)	(0.763)	(0.707)	(0.645)	(0.432)
Observations	44	44	37	36	44	44	44
R-squared	0.39	0.36	0.41	0.45	0.36	0.34	0.31

Notes: Results of regressing the four-quarter sum of different GDP growth (for the 10 % quantile) responses to a one-standard-deviation U.S. financial shock (panel 1) or monetary policy shock (panel 2) on country characteristics. Columns report alternative specifications with: (1) de jure measure of financial openness (Chinn and Ito, 2008); (2) financial openness measured by cross-border banking claims/GDP; (3) excluding financial centers as defined by Lane and Milesi-Ferretti (2018) (BEL, CHE, GBR, HKG, IRL, LUX, NLD, and SGP); (4) including the share of countries' non-US trade invoiced in US dollar (Boz et al., 2020); (5) exchange rate regime adjusted for euro area members according to PPP-GDP weight; and (6) GDP response from an alternative path for future quantiles in which, after two quarters following the shock, the quantile value for GDP growth switches from 10 % (90 %) to 50 %; (7) with response from quantile local projections after four quarters as the dependent variable. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively. LTV denotes loan to value.

Data availability

Data will be made available on request.

References

Abbate, A., Eickmeier, S., Lemke, W., Marcellino, M., 2016. The changing international transmission of financial shocks: evidence from a classical time-varying FAVAR. J. Money Credit Bank. 48, 573–601.

Adrian, T., Boyarchenko, N., Giannone, D., 2019. Vulnerable growth. Am. Econ. Rev. 109, 1263–1289.

Adrian, T., Duarte, F., Liang, N., Zabczyk, P., 2020. NKV: a new Keynesian model with vulnerability. AEA Papers Proc. 110, 470–476.

Adrian, T., Grinberg, F., Liang, N., Malik, S., Yu, J., 2022. The term structure of growth-at-risk. Am. Econ. J. Macroecon. 14, 283–323.

Aikman, D., Bridges, J., Hoke, S.H., O'Neill, C., Raja, A., 2019. Credit, capital and crises: a GDP-at-risk approach. Bank of England Staff Working Paper 824.

Alam, Z., Alter, A., Eiseman, J., Gelos, G., Kang, H., Narita, M., Nier, E., Wang, N., 2019. Digging deeper – evidence on the effects of macroprudential policies from a new database. IMF Working Papers 19/66.

Arbatli-Saxegaard, E.C., Furceri, D., Dominguez, P.G., Ostry, J.D., Peiris, S.J., 2024. Spillovers from us monetary policy: role of policy drivers and cyclical conditions. J. Int. Money Finance 143, 103053.

Bekaert, G., Mehl, A., 2019. On the global financial market integration "swoosh" and the trilemma. J. Int. Money Finance 94, 227–245.

Bénétrix, A., Gautam, D., Juvenal, L., Schmitz, M., 2019. Cross-border currency exposures. New evidence based on an enhanced and updated dataset. IMF Working Paper 19/299.

Berkmen, S.P., Gelos, G., Rennhack, R., Walsh, J.P., 2012. The global financial crisis: explaining cross-country differences in the output impact. J. Int. Money Finance 31, 42–59.

Bianchi, J., 2011. Overborrowing and systemic externalities in the business cycle. Am. Econ. Rev. 101, 3400–3426.

Blomqvist, N., 1950. On a measure of dependence between two random variables. Ann. Math. Stat. 21, 593–600.
Born, A., Enders, Z., 2019. Global banking, trade, and the international transmission of the great recession. Econ. J. 129, 2691–2721.
Boz, E., Casas, C., Georgiadis, G., Gopinath, G., Le Mezo, H., Mehl, A., Nguyen, T., 2020. Patterns in invoicing currency in global trade. IMF Working Paper 20/126.
Brownlees, C., Souza, A.B., 2021. Backtesting global growth-at-risk. J. Monet. Econ. 118, 312–330.
Bruno, V., Shin, H., 2015. Capital flows and the risk-taking channel of monetary policy. J. Monet. Econ. 71, 119–132.
Buch, C.M., Eickmeier, S., Prieto, E., 2014. Macroeconomic factors and microlevel bank behavior. J. Money Credit Bank. 46 (4), 715–751.
Cecchetti, S.G., 2008. Measuring the macroeconomic risks posed by asset price booms. In: Campbell, J.Y. (Ed.), Asset Prices and Monetary Policy. University of Chicago Press, pp. 9–43.
Cesa-Bianchi, A., Ferrero, A., Rebucci, A., 2018. International credit supply shocks. J. Int. Econ. 71, 219–237.
Chavleishvili, S., Manganelli, S., 2024. Forecasting and stress testing with quantile vector autoregression. J. Appl. Econ. 39 (1), 66–85.
Chinn, M.D., Ito, H., 2006. What matters for financial development? Capital controls, institutions, and interactions. J. Dev. Econ. 81, 163–192.
Chinn, M.D., Ito, H., 2008. A new measure of financial openness. J. Compar. Policy Anal. 10, 309–322.
Chuliá, H., Garrón, I., Uribe, J.M., 2024. Vulnerable funding in the global economy. J. Bank. Finance 169, 107314.
Dahlhaus, T., Vasishtha, G., 2021. Reprint: monetary policy news in the US: effects on emerging market capital flows. J. Int. Money Finance 114, 102403 (Special Issue "Monetary Policy under Global Uncertainty").
De Nicoló, G., Lucchetta, M., 2017. Forecasting tail risks. J. Appl. Econ. 32, 159–170.
Dedola, L., Lippi, F., 2005. The monetary transmission mechanism: evidence from the industries of five OECD countries. Eur. Econ. Rev. 49, 1543–1569.
Dedola, L., Rivolta, G., Stracca, L., 2017. If the Fed sneezes, who catches a cold? J. Int. Econ. 108, S23–S41.
Degasperi, R., Hong, S., Ricco, G., 2020. The global transmission of US monetary policy. CEPR Discussion Paper 14533.
Edwards, S., 2004. Financial openness, sudden stops, and current-account reversals. Am. Econ. Rev. 94 (2), 59–64.
Eichengreen, B., Gupta, P., 2016. Managing sudden stops. World Bank Policy Research Working Paper 7639.
Eickmeier, S., Gambacorta, L., Hofmann, B., 2014. Understanding global liquidity. Eur. Econ. Rev. 68, 1–18.
Eickmeier, S., Ng, T., 2015. How do credit supply shocks propagate internationally? A GVAR approach. Eur. Econ. Rev. 74, 128–145.
Emter, L., Schmitz, M., Tirpák, M., 2019. Cross-border banking in the EU since the crisis: what is driving the great retrenchment? Weltwirtsch. Arch. 155, 287–326.
Farhi, E., Werning, I., 2016. A theory of macroprudential policies in the presence of nominal rigidities. Econometrica 84, 1645–1704.
Favara, G., Gilchrist, S., Lewis, K.F., Zakrajsek, E., 2016. Updating the recession risk and the excess bond premium. FEDS Notes. Board of Governors of the Federal Reserve System, Washington. doi:https://doi.org/10.17016/2380-7172.1836.
Fernald, J.G., 2014. A quarterly, utilization-adjusted series on total factor productivity. FRBSF Working Paper 2012–19.
Fink, F., Schüler, Y.S., 2015. The transmission of US systemic financial stress: evidence for emerging market economies. J. Int. Money Finance 55, 6–26.
Fleming, J.M., 1962. Domestic financial policies under fixed and under floating exchange rates. Imf Staff Papers 9, 369–380.
Fratzscher, M., Lo Duca, M., Straub, R., 2017. On the international spillovers of US quantitative easing. Econ. J. 128, 330–377.
Gaechter, M., Geiger, M., Hasler, E., 2023. On the structural determinants of growth-at-risk. Int. J. Cent. Bank. 19, 251–293.
Gallant, A., Rossi, P., Tauchen, G., 1993. Nonlinear dynamic structures. Econometrica 61, 871–907.
Gambacorta, L., Hofmann, B., Peersman, G., 2014. The effectiveness of unconventional monetary policy at the zero lower bound: a cross-country analysis. J. Money Credit Bank. 46, 615–642.
Gilchrist, S., Zakrajsek, E., 2012. Credit spreads and business cycle fluctuations. Am. Econ. Rev. 102, 1692–1720.
Gourinchas, P.-O., Obstfeld, M., 2012. Stories of the twentieth century for the twenty-first. Am. Econ. J. Macroecon. 4, 226–265.
Guerrieri, L., Iacoviello, M., 2017. Collateral constraints and macroeconomic asymmetries. J. Monet. Econ. 90, 28–49.
Helbling, T., Huidrom, R., Kose, M.A., Otrok, C., 2011. Do credit shocks matter? A global perspective. Eur. Econ. Rev. 55, 340–353.
Ilut, C.L., Schneider, M., 2014. Ambiguous business cycles. Am. Econ. Rev. 104 (8), 2368–2399.
Ilzetzki, E., Reinhart, C.M., Rogoff, K.S., 2019. Exchange arrangements entering the twenty-first century: which anchor will hold? Q. J. Econ. 134, 599–646.
Jorda, O., Schularick, M., Taylor, A.M., 2013. When credit bites back. J. Money Credit Bank. 45, 3–28.
Kim, S., 2001. International transmission of U.S. monetary policy shocks: evidence from VAR's. J. Monet. Econ. 48, 339–372.
Klein, M.W., Shambaugh, J.C., 2015. Rounding the corners of the policy trilemma: sources of monetary policy autonomy. Am. Econ. J. Macroecon. 7, 33–66.
Koenker, R., Bassett, G., 1990. M estimation of multivariate regressions. J. Am. Stat. Assoc. 85, 1060–1068.
Koenker, R., Machado, J., 1999. Goodness of fit and related inference processes for quantile regression. J. Am. Stat. Assoc. 94, 1296–1310.
Koop, G., Pesaran, M., Potter, S., 1996. Impulse response analysis in nonlinear multivariate models. J. Econom. 74, 119–147.
Korinek, A., Simsek, A., 2016. Liquidity trap and excessive leverage. Am. Econ. Rev. 106, 699–738.
Lane, P.R., Milesi-Ferretti, G.M., 2007. The external wealth of nations mark II: revised and extended estimates of foreign assets and liabilities, 1970–2004. J. Int. Econ. 73, 223–250.
Lane, P.R., Milesi-Ferretti, G.M., 2018. The external wealth of nations revisited: international financial integration in the aftermath of the global financial crisis. IMF Econ. Rev. 66, 189–222.
Lane, P.R., Shambaugh, J.C., 2010. Financial exchange rates and international currency exposures. Am. Econ. Rev. 100 (1), 518–540.
Loria, F., Matthes, C., Zhang, D., 2025. Assessing macroeconomic tail risk. Econ. J. 135 (665), 264–84.
Mackowiak, B., 2007. External shocks, U.S. monetary policy and macroeconomic fluctuations in emerging markets. J. Monet. Econ. 54, 2512–2520.
Metiu, N., 2016. How does the stock market respond to changes in bank lending standards? Econ. Lett. 144, 92–97.
Metiu, N., 2021. Anticipation effects of protectionist U.S. trade policies. J. Int. Econ. 133, 103536.
Metiu, N., Hilberg, B., Grill, M., 2016. Credit constraints and the international propagation of US financial shocks. J. Bank. Finance 72, 67–80.
Mian, A., Sufi, A., Verner, E., 2017. Household debt and business cycles worldwide. Q. J. Econ. 132, 1755–1817.
Miranda-Agrippino, S., Rey, H., 2020. US monetary policy and the global financial cycle. Rev. Econ. Stud. 87 (6), 2754–2776.
Miranda-Agrippino, S., Ricco, G., 2021. The transmission of monetary policy shocks. Am. Econ. J. Macroecon. 13 (3), 74–107
Montes-Rojas, G., 2019. Multivariate quantile impulse response functions. J. Time Ser. Anal. 40, 739–752.
Mundell, R.A., 1960. The monetary dynamics of international adjustment under fixed and flexible exchange rates. Q. J. Econ. 74, 227–257.
Obstfeld, M., 2021. Trilemmas and tradeoffs: living with financial globalization. In: Davis, S., Robinson, E.S., Yeung, B. (Eds.), The Asian Monetary Policy Forum, Insights for Central Banking, World Scientific Book Chapters, Chapter 2. World Scientific Publishing, pp. 16–84.
Obstfeld, M., Ostry, J.D., Qureshi, M.S., 2019. A tie that binds: revisiting the trilemma in emerging market economies. Rev. Econ. Stat. 101, 279–293.
Obstfeld, M., Shambaugh, J.C., Taylor, A.M., 2005. The trilemma in history: tradeoffs among exchange rates, monetary policies, and capital mobility. Rev. Econ. Stat. 87, 423–438.
Pesaran, M.H., Smith, R., 1995. Estimating long-run relationships from dynamic heterogeneous panels. J. Econom. 68, 79–113.
Petrella, L., Raponi, V., 2019. Joint estimation of conditional quantiles in multivariate linear regression models with an application to financial distress. J. Multivar. Anal. 173, 70–84.
Plagborg-Møller, M., Reichlin, L., Ricco, G., Hasenzagl, T., 2020. When is growth at risk? In: Brookings Papers on Economic Activity. Spring, pp. 167–229.
Ramey, V., 2016. Chapter 2—macroeconomic shocks and their propagation. In: Volume 2 of Handbook of Macroeconomics. Elsevier, pp. 71–162.
Reinhart, C.M., Rogoff, K.S., 2014. Recovery from financial crises: evidence from 100 episodes. Am. Econ. Rev. 104, 50–55.
Rey, H., 2015. Dilemma not trilemma: the global financial cycle and monetary policy independence. NBER Working Paper 21162.
Schüler, Y.S., 2020. The impact of uncertainty and certainty shocks. Bundesbank Discussion Paper 2020/14.
Wei, Y., 2008. An approach to multivariate covariate-dependent quantile contours with applications to bivariate conditional growth charts. J. Am. Stat. Assoc. 103, 397–409.
Yu, K., Moyeed, R.A., 2001. Bayesian quantile regression. Stat. Probab. Lett. 54, 437–447.